

UNIVERSITY OF MEDICAL SCIENCES, ONDO
POST UNIFIED TERTIARY MATRICULATION SCREENING

2015/2016

Instructions

Section 1 English

Read the passage below and answer questions that follow

The rich countries of the world, Europe and North America in particular, became rich for several reasons. They were developing at a time when there were still large areas of unexplored and sparsely inhabited land in the world they discovered; and they then began to exploit whatever resources they found.

At the same time, at home, the industrial revolution was going on and the raw materials obtained from the countries they colonised were useful in developing new industries. The products of these industries provided an incentive for people in the mother country to invent more and work harder. The products, for example guns, were used in subduing the peoples of other parts of the world, or in enticing them into the economic orbit of the industrialising countries.

The people of the countries of the world now called developing countries were impressed by the technology of the Europeans. Not without some justification, they connected the wealth of Europeans with their technology alone, being unaware of the other factors contributing to their progress. North America developed, riding on the back of European industrialisation and colonisation. Today America and Europe have forged ahead, so that the *per capita* income in the USA may be ten times that in the poorer countries.

The latter now tend to feel that their way of development and future economic sufficiency must be along the same route as that taken by the wealthier countries. This may not necessarily be so, as the circumstances in which they are developing are very different.

Explain the meaning of the following words in the passage

1. An incentive
 - a. A case to handle
 - b. A motive for them
 - c. Reason for working harder+++
 - d. Something to remind them
2. Enticing
 - a. Forcing
 - b. Allowing
 - c. Begging
 - d. Persuading+++
3. Orbit of
 - a. Sphere of influence+++
 - b. Usual progression
 - c. Revolving round
 - d. Member of

4. What figure of speech is used in “riding on the back of”
 - a. Simile
 - b. Metaphor+++
 - c. Personification
 - d. Irony
5. In the writers opinion the way to development of the poorer countries must ----- as that taken by wealthier countries.
 - a. Be along the same route
 - b. Take a different route+++
 - c. Should walk side by side
 - d. Should rely on their resources

Read the following passage and select the best option that fills the corresponding gap in the text.

With the decline in sales of manufactured goods at home and abroad, widespread ---- 6 --- is likely to affect many firms. The motor trade employs the largest sector of the working -----7 ---- and will probably be first affected. The government has therefore set up a ---8---- to examine ways and means of ----9--- -- new industries in areas where previously labour was not available. Unemployment is the most serious of all-----10 ----- problems

6.
 - a. promotion
 - b. redundancy+++
 - c. surplus
 - d. abundance
7.
 - a. population+++
 - b. congregation
 - c. crowd
 - d. denomination
8.
 - a. commitment
 - b. profession
 - c. committee+++
 - d. commune
9.
 - a. searching
 - b. evolving
 - c. developing +++
 - d. growing
10.
 - a. industrious
 - b. industrial+++

- c. employment
- d. workers

INSTRUCTION: Fill the gaps in the sentences below by choosing the word which is most **nearly opposite in meaning** to the one underlined in each sentence, and which also best completes the sense of the whole.

11. Before he started playing football he used to be overweight for his age and height; now he is actually -----
 - a. Light-weight
 - b. Underweight+++
 - c. light
 - d. under-balanced

12. Last week it was doubtful if Kola would recover from his illness, but now it is almost -----that he will.
 - a. Assured
 - b. Factual
 - c. Genuine
 - d. Certain+++

13. The questions in this exercise require----- answers, but the answers to the composition questions have to be approached subjectively.
 - a. Objective +++
 - b. Factual
 - c. Opposite
 - d. Realistic

14. His invention may work in theory but it is not-----
 - a. Application
 - b. Effective
 - c. Practical +++
 - d. Convenient

15. At first they thought the car belonged to the enemy, but to their relief it had already been captured by their-----
 - a. Comrades
 - b. Associates
 - c. Collaborators
 - d. Allies+++

16. I can't believe Femi would throw the ring into the river deliberately .He must have done it----
 - a. Incidentally
 - b. Spontaneously
 - c. Accidentally+++
 - d. Crazyly.

17. The government is going to -----all the old buildings in this area and have new ones erected.
- Abolish
 - End
 - Terminate
 - Demolish+++

Section 2 – General paper

18. Simplify $\sqrt{27} \times \sqrt{50}$
- $8\sqrt{5}$
 - $15\sqrt{6}$ +++
 - $8\sqrt{6}$
 - $15\sqrt{5}$
19. Given that $\log 2 = 0.30103$ and $\log 3 = 0.47712$, without using tables, calculate $\log 6$.
- 0.09062
 - 0.60206
 - 0.77815+++
 - 0.14363
20. 6 notebooks cost ~~N~~300.00, how many books can be bought with ~~N~~450.00.
- 8
 - 7
 - 9+++
 - 10
21. Factorize $5m^2 - 80$
- $5(m + 4)^2$
 - $5(m^2 - 16)$
 - $5(m + 4)(m - 4)$ +++
 - $16(m^2 - 5)$
22. Without using tables or a calculator, calculate 12.9×0.54
- 5.966
 - 6.966+++
 - 6.968
 - 5.968
23. Express 2.4 Kg in g
- 240
 - 2400+++
 - 24
 - 480
24. Which of the following is a non-luminous object?
- Sun
 - Moon+++
 - Stars
 - Lighted candle

25. If the velocity of light in air is $5.0 \times 10^8 \text{ ms}^{-1}$, find the velocity of light in a medium whose refractive index is 2.0
- $1 \times 10^{-9} \text{ ms}^{-1}$
 - $2.5 \times 10^8 \text{ ms}^{-1}+++$
 - $2.5 \times 10^{-8} \text{ ms}^{-1}$
 - $1 \times 10^9 \text{ ms}^{-1}$
26. In which medium do sound waves have the greatest speed
- Vacuum
 - Air
 - Water
 - Steel+++
27. What is the kinetic energy of a boy of mass 50kg running with a velocity of 3 ms^{-1} ?
- 7500 J
 - 150 J
 - 225 J+++
 - 450 J
28. What is the potential energy of a swimmer of mass 65 kg at a height of 15 m above the swimming pool? ($g = 9.8 \text{ ms}^{-1}$)
- 9408 J
 - 9555 J+++
 - 4777.5 J
 - 4704 J
29. Convert 200 Kelvin to $^{\circ}\text{C}$.
- 70°C
 - 73°C
 - -70°C
 - $-73^{\circ}\text{C}+++$
30. Why does water wet its glass container and mercury does not?
- Water is a more viscous liquid than mercury.
 - Mercury is a more viscous liquid than water.
 - The adhesive forces between water and glass molecules is more than the cohesive forces between water molecules+++
 - The adhesive forces between mercury and glass molecules is more than the cohesive forces between mercury molecules.
31. When a brass metal is heated, it expands because its molecules
- Vibrate with smaller amplitude
 - Vibrate with greater amplitude+++
 - Increase their speed of random motion
 - Decrease their speed of random motion.
32. When oxygen gas is heated, it expands because its molecules

- a. Vibrate with smaller amplitude
 - b. Vibrate with greater amplitude
 - c. Increase their speed of random motion+++
 - d. Decrease their speed of random motion.
33. Why do leaves of plants appear green?
- a. Leaves use green light in photosynthesis
 - b. Leaves absorb green light
 - c. Leaves reflect green light+++
 - d. Leaf thickness is smaller than the wavelength of green light
34. How much NaCl would be required to prepare normal saline (0.9% NaCl) in a 250 ml volumetric flask?
- a. 1.8 g
 - b. 13.2 g
 - c. 2.25 g+++
 - d. 9.0.
35. A sample of grape juice has pH of 3.80. What is the molar concentration of hydrogen ion in the juice?
- a. 1.58×10^4
 - b. 1.58×10^2
 - c. 1.58×10^{-2}
 - d. 1.58×10^{-4} +++
36. Which of the following are products of hydrolysis of lactose?
- a. Glucose and fructose
 - b. Glucose and galactose+++
 - c. Glucose and mannose
 - d. Glucose and maltose
37. At 40°C, the volume of a gas was 1.2 m³, at what temperature will the gas occupy a volume of 2.5 m³ when heated under constant pressure?
- a. 313K
 - b. 652K+++
 - c. 320K
 - d. 330K
38. In a 500 ml volumetric flask, what quantity of Na₂CO₃ would be required to prepare a 2.5M solution? [Na = 23; C = 12; O = 16].
- a. 106 g
 - b. 212 g
 - c. 265 g
 - d. 132.5 g+++

39. The relationship between Gibbs' free energy (G), enthalpy (H), and entropy (S) is represented as follows
- $\Delta H = T\Delta S - \Delta G$
 - $\Delta G = T\Delta S - \Delta H$
 - $\Delta G = \Delta H - T\Delta S$
 - $\Delta S = T\Delta H - \Delta G$
40. Which of the following signifies that a reaction is at equilibrium
- Positive ΔG
 - Negative ΔG
 - Positive ΔS
 - Zero ΔG
41. 7.8g of NaNO_3 was dissolved in 100 ml of water at 25°C . Calculate the solubility of the salt in mol dm^{-3} . [Na = 23; N = 14; O = 16].
- 0.918
 - 0.0918
 - 22.94
 - 273.46
42. Which of the following can be regarded as the strongest bonds between atoms
- Covalent bonds
 - Electrostatic bonds
 - Hydrogen bonds
 - Van der Waals forces
43. A gas occupies a volume of 30 cm^3 at a temperature of 26°C and a pressure of 75 cmHg. What volume will it occupy at 70°C and pressure of 120 cmHg.
- 16.34 cm^3
 - 31.20 cm^3
 - 50.48 cm^3
 - 21.51 cm^3
44. Which of the following is not a function of the human blood
- Defense
 - Transport
 - Hemostasis
 - Cellular respiration
45. Which of these cell organelles is responsible for cellular respiration
- Golgi bodies
 - Nucleus
 - Mitochondria
 - Ribosomes
46. In blood transfusion a universal donor has the following blood group
- AB negative

- b. AB positive
 - c. O negative+++
 - d. O positive
47. The following animal has a heterodont dentition
- a. Tilapia
 - b. Toad
 - c. Sheep+++
 - d. Lizard
48. The following is not a plant hormones
- a. Gibberillins
 - b. Ethylene
 - c. Cytokines+++
 - d. Absciscic acid
49. Which of the following is not a communicable disease
- a. Tuberculosis
 - b. Measles
 - c. Diabetes+++
 - d. Common cold
50. The following hormone is responsible for lowering blood glucose levels
- a. Glucagon
 - b. Thyroxine
 - c. Insulin+++
 - d. Adrenaline

2017/2018 POST UNIFIED TERTIARY MATRICULATION SCREENING

DAY 1

- 1 A car travels 90 m due north in 15 s. then the car turns around and travels 40 m due south in 5 s. what is the magnitude of the average velocity of the car during this 20 s interval?
 a. **2.5m/s Ans**
 b. 5.0m/s
 c. 6.5m/s
 d. 7.0m/s
- 2 A worker uses a rope to pull a cart; the worker's pull on the handle of the cart can best be described as a force having
 a. Magnitude only
 b. Direction only
 c. **Both magnitude and direction Ans**
 d. Magnitude or direction
- 3 When a neutral metal sphere is charged by contact with a positively charged glass rod, the sphere

a. **Loses electrons** **Ans**

b. Gains electrons

c. Loses protons

d. Gains protons

4 Sound travels in air as:

a. **longitudinal waves** **Ans**

b. transverse waves

c. electromagnetic waves

d. matter waves

5 Which of the following is a scalar quantity?

a. the braking force needed to stop a car

b. the effort needed to hammer a nail into wood

c. **the heat needed to boil some water** **Ans**

d. the thrust needed to lift a rocket off the ground

6 Using a stopwatch, which of the following is correct to determine the time for one oscillation of a pendulum?

a. **Time for 20 oscillations and divide by 20** **Ans**

b. Time for 20 oscillations and multiply by 20.

c. Time for one oscillation.

d. None of the above.

7 What does Newton's third law of motion say?

a. Acceleration is a result of resultant force acting on a body.

b. Resultant force is always equal but opposite to the force acting on a body.

c. For every action there is an equal but opposite reaction.

d. **The forces acting on a body are always equal and opposite.** **Ans**

8 An object Q is moving in a circle around a point R. A force F keeps it moving in the circle. If the force F suddenly disappears, Q will

a. **goes off in a straight line.** **Ans**

b. moves towards R.

c. moves away from R.

d. moves away in a curved path.

9 A heater marked 50W boils water for 50s, if the specific latent heat of evaporation is $5 \times 10^5 \text{ J kg}^{-1}$ the mass of water that evaporates is

a. 50 kg

b. **0.005 kg** **Ans**

c. 0.1 kg

d. 0.05 kg

10 The input voltage of a transformer with 300 turns of wire in the primary coil is 100V, if the output voltage of the transformer is 10V, what is the number of turns in the secondary coil?

a. 10

b. **30** **Ans**

c. 20

d. 15

11 The speed of light in air is $3 \times 10^8 \text{ ms}^{-1}$, if its speed in glass is $1.82 \times 10^8 \text{ ms}^{-1}$, the refractive index of the glass is

a. 1.82

b. 3

c. 1.65

Ans

d. 6

12 What is the latent heat of fusion of ice if the heat required to change 10kg of ice at 0°C into water at the same temperature is 800J?

a. 8 kg^{-1}

b. 10 kg^{-1}

c. **80 Jkg^{-1} Ans**

d. 800 Jkg^{-1}

13 It is easier to turn a tight nut and bolt by using a longer spanner because a longer spanner gives

a. more weight.

b. more friction

c. **bigger turning moment. Ans**

d. bigger pressure.

14 All the following metals are solid except

a. Cadmium

b. Lead

c. **Mercury Ans**

d. Sodium

15 Carbon has ----- allotropes

a. One

b. **Two Ans**

c. Three

d. Four

16 The catalysts for Contact process and Haber process are ----- and ----- respectively

a. Iron and Nickel

b. Iron and Vanadium oxide

c. **Vanadium oxide and Iron Ans**

d. Nickel and Iron

17 An example of acidic salt is -----

a. NaCl

b. **KH_2PO_4 Ans**

c. $\text{Zn}(\text{OH})\text{Cl}$

d. $\text{KCr}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$

18 In a sample containing 100 atoms of X, 70 were found to be ^9X while 30 were ^{11}X isotopes. Determine the average relative atomic mass

- a. 6.3g
 - b. **9.6g** **Ans**
 - c. 10.4g
 - d. 3.3g
- 19 The electrode in which metallic radicals migrate to during electrolysis is termed -----
- a. anode
 - b. **cathode** **Ans**
 - c. glass electrode
 - d. selective ion electrode
- 20 A part of an organic compound that determines its chemical property is called-----
- a. homologues series
 - b. empirical formula
 - c. **functional group** **Ans**
 - d. isomerism
- 21 The scope of organic Chemistry does not include
- a. carbon and its compounds
 - b. animal products
 - c. **transition metals** **Ans**
 - d. plant products
- 22 The reaction $2\text{C}_2\text{H}_2 + 5\text{O}_2$ is an example of
- a. oxidation reaction
 - b. addition reaction
 - c. **combustion** **Ans**
 - d. oxidation process
- 23 The degree of unsaturation of edible oil can be known by its -----
- a. saponification value
 - b. acid value
 - c. **iodine value** **Ans**
 - d. esterification value
- 24 The identification of various species present in a mixture of species without bothering on the quantity is called.....
- a. analysis
 - b. quantitative analysis
 - c. **qualitative analysis** **Ans**
 - d. protolysis
- 25 Preliminary test involves all the followings except
- a. physical appearance
 - b. **chemical reaction** **Ans**
 - c. solubility test
 - d. flame test

- 26 A solution does not change the colour of red litmus paper when dipped in it. The solution must likely be.....
- a. acidic
 - b. basic
 - c. neutral
 - d. acidic or neutral Ans**
- 27 Heterotrophic mode of life with extracellular digestion is a feature of
- a. Lichen
 - b. Fungi Ans**
 - c. Algae
 - d. Cyanobacteria
- 28 Lack of structures to represent the leaf, stem and root are unique to
- a. Liverwort
 - b. Moss
 - c. Algae Ans**
 - d. Mycorrhiza
- 29 Which of these transports water and minerals salts in pteridophytes
- a. Phloem
 - b. Rhizoid
 - c. Root
 - d. Xylem Ans**
- 30 Seed producing plants are collectively called
- a. Gymnosperm Ans**
 - b. Spermatophyte
 - c. Angiosperm
 - d. Pteridophytes
- 31 Which of the following shows highest evolutionary advancement
- a. Moss
 - b. Fern
 - c. Spirogyra
 - d. Pine Ans**
- 32 Complete metamorphosis occurs in one of the following insects
- a. Cockroach
 - b. Butterfly Ans**
 - c. Beetle
 - d. Aphid

33 The reverse process of this equation $6\text{CO}_2 + 6\text{H}_2\text{O} \dots\dots\dots \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$

- a. Translocation
- b. Digestion
- c. Respiration Ans**
- d. Nutrition

34 One of these is not a fat soluble vitamin

- a. Vitamin A
- b. Vitamin C Ans**
- c. Vitamin E
- d. Vitamin K

35 Benedict's solution is used to test for

- a. Starch
- b. Proteins
- c. Fats
- d. Reducing sugars Ans**

36 The cell structure located near the nucleus used in cell division is

- a. Centriole Ans**
- b. Lysosome
- c. Endoplasmic reticulum
- d. Ribosome

37 The building up biological process is called

- a. Catabolism
- b. Isomerism
- c. Anabolism Ans**
- d. Metabolism

38 The knot of capillaries in the Bowman's capsule is called

- a. Cortex
- b. Medulla
- c. Glomerulus Ans**
- d. Pelvis

39 Which of the following organisms moves with the aid of flagella?

- a. Amoeba
- b. Paramecium
- c. Trypanosome Ans**
- d. Hydra

40 The portion of the blood cells responsible for blood clotting is

- a. Erythrocytes
- b. Leucocytes
- c. Platelets Ans**
- d. Monocytes

41 Why is there shortage of meter? **Read the passages below and answer the questions that follow:**
Nigeria has a troubled power sector which is however undergoing reforms that, hopefully, should usher in era of private sector dominance in terms of ownership and management of generation and distribution. In this transitory phase, it is crucial to lay a foundation for local sourcing of vital inputs and make it the norm for the industry. That is the only way to prevent a replication of the trend in the oil and gas sub-sector, where capital flight is as much as \$ 16 billion per annum, as revealed by the Petroleum Technology Development Fund. The PHCN could legitimately insist on quality and standards in its procurement of equipment and materials, but what is playing out is an institutional orientation, evinced in several pursuits of the PHCN that are contemptuous of Nigeria's aspiration for local content development. In 1998, the defunct National Electric Power Authority defied a directive by the then Minister of Power and steel, Alhaji Bashir Dalhatu, to source its costable components from Nigerian foundries, some of which are of high standards. Actuated by a realization of idle capacity in the foundries and the need to conserve foreign exchange, the Minister applied appropriate pressure on the management of NEPA but found his efforts frustrated. The acute shortage or inefficiency in the metering programme originated from the fact that the only local manufacturer then, the Federal Government-owned electricity Meter Company in Zaria, Kaduna State, established in 1976, but sold to Dantata Investments Limited in December 2002, is not performing. (Adapted from *The Punch*, Wednesday, May 25, 2011. Pg. 18)

- a. Non-performance of PHCN
- b. Non-performance of NEPA
- c. Non-performance of the Federal Government-owned electricity meter company
- d. Non-performance of Dantata Investment Ltd. Ans**

42 According to the passage, the solution to the problem in the power sector lies in:

- a. Sourcing for materials locally Ans**
- b. Getting costable components
- c. Obeying the minister
- d. Getting \$ 16 billion per annum

43 A suitable title for this passage is

- a. Nigeria and her problems
- b. NEPA versus PHCN
- c. Electricity meter
- d. Problems in the power sector Ans**

44 Ladies and gentlemen, the worse may be over, but we are not out of the woods yet. This means that we are **Choose the interpretation that you consider most appropriate for this sentences:**

- a. still in the forest
- b. not yet ready to celebrate our success
- c. likely to encounter more difficulties Ans**
- d. not sure of ourselves

45 Surprisingly, Musa turned.....the job offer ***Choose the word or group of words that best completes the following sentence:***

- a. away
- b. in
- c. out
- d. down Ans**

46 We have no choice..... to do what they demanded. ***Choose the word or group of words that best completes the following sentence:***

- a. except
- b. than
- c. unless
- d. but Ans**

47 In this matter, we must be prepared to tolerate human frailty. ***Choose the option that is nearest in meaning to the word underlined:***

- a. inquisitiveness
- b. weakness Ans**
- c. profligacy
- d. innovation.

48 I detest his lackadaisical attitude. ***Choose the option that is nearest in meaning to the word underlined:***

- a. carefree Ans**
- b. lazy
- c. supercilious
- d. disloyal.

49 Peter always approaches issues optimistically. ***Choose the option that is opposite in meaning to the word underlined.***

- a. realistically
- b. pessimistically Ans**
- c. carelessly
- d. unrealistically.

50 Why should Dixon be evasive in answering the question? ***Choose the option that is opposite in meaning to the word underlined.***

- a. indirect
- b. direct Ans**
- c. guessing
- d. sure.

DAY 2

- 1 How far will a brick starting from rest fall freely in 3 s?
 - a. 15 m

- b. 29 m
- c. 44 m Ans**
- d. 88 m

2 If the sum of all the forces acting on a moving object is zero, the object will

- a. Slow down and stop
- b. Change the direction of its motion
- c. Accelerate uniformly
- d. Continue moving with constant velocity Ans**

3 Focal length of plane mirror is:

- a. zero
- b. one
- c. infinity Ans**
- d. depends upon quality of the mirror

4 The device used for measuring potential difference is:

- a. voltmeter Ans**
- b. ammeter
- c. galvanometer
- d. thermometer

5 Which of the following is measured using a micrometer?

- a. current
- b. length Ans**
- c. resistance
- d. weight

6 What is the correct unit for density?

- a. m^3/kg
- b. kg^3/m^2
- c. kg/m^3 Ans**
- d. kg/m^2

7 Frictional force is necessary for

- a. a ball to bounce off a wall.
- b. a car to accelerate. Ans**
- c. a rocket to be launch.
- d. a spring to vibrate.

8 Which statement about frictional force is true?

- a. It is required to start a motion. Ans**
- b. It acts in the direction of the motion.
- c. Its value is equal to the force producing the motion.
- d. It can be reduced by increasing the force producing the motion.

9 If the maximum resistance and current of an electric iron is $62.5\ \Omega$, 4A. The electric iron is rated

- a. 40W, 220V

- b. 160W, 250V
- c. 1000W, 250V Ans**
- d. 400W, 120V

10 Two lamps rated 220V each are connected in series. If the total power dissipated in both lamps is 80W. What is the rated power of the lamps?

- a. 10W
- b. 20W
- c. 80W
- d. 40W Ans**

11 A machine gun with a mass of 5kg fires a bullet at a speed of 100ms^{-1} . The recoil speed of the machine gun is 1ms^{-1} , what is the mass of the bullet?

- a. 5g
- b. 50g Ans**
- c. 10g
- d. 20g

12 If the current in each of the three 5Ω resistors connected in parallel is 12A, what is the potential difference applied across the combination?

- a. 40V
- b. 120V
- c. 60V Ans**
- d. 24V

13 Which of the following is not a transverse wave?

- a. light waves
- b. string waves
- c. electromagnetic waves
- d. sound waves Ans**

14 Which of the following metals react reversibly with steam?

- a. Sodium
- b. Iron Ans**
- c. Calcium
- d. Aluminium

15 Which ion below has a noble gas electron configuration?

- a. B^{2+}
- b. Be^{2+} Ans**
- c. Li^{2+}
- d. N^{2-}

16 Which of the following metals cannot be extracted by electrolysis?

- a. sodium
- b. calcium
- c. tin Ans**
- d. magnesium

17 ----- described that atom is built of three sub particles which are proton, electron and neutron

- a. John Dalton
- b. Lord Rutherford Ans**
- c. Michael Faraday
- d. J.J. Thompson

18 When the energy of product of a reaction is greater than that of reactant of the reaction, the reaction is said to be -----

- a. isothermic
- b. endothermic Ans**
- c. exothermic
- d. octothermic

19 A change that has a tendency to occur without needing to be driving by an external influence is -----

- a. non-spontaneous
- b. enthalpy
- c. spontaneous Ans**
- d. entropy

20 The carbon atom in propyne is.....hybridization

- a. sp^3
- b. sp Ans**
- c. sp^2
- d. sp^3d

21 The final product in the halogenation and substitution of methane is

- a. monochloromethane
- b. tetrachloromethane Ans**
- c. chloroform
- d. dichloromethane

22 All these are organic compounds/molecules except

- a. carbon (IV) oxide Ans**
- b. methanol
- c. marsh gas
- d. olefin

23 The process of combining dicarboxylic acids and diols to form polyester is called -----

- a. addition polymerization
- b. condensation polymerization Ans**
- c. aldol condensation
- d. free radical polymerization

24 A solid substance is dissolved with a mineral acid in a test tube and the test tube is warmed when touched. The reaction that occurred is known as.....

- a. neutralization
- b. redox
- c. endothermic
- d. exothermic Ans**

25 One of the following salts has no residue after heating

- a. lead chloride
- b. sodium chloride
- c. **ammonium chloride** **Ans**
- d. iron chloride

26 Which of these following salts is yellow when hot but white when cold?

- a. lead salt
- b. iron salt
- c. **zinc salt** **Ans**
- d. copper salt

27 An example of a unicellular, motile alga is

- a. **Chlorella** **Ans**
- b. *Chlamydomonas*
- c. *Ulva*
- d. Spirogyra

28 The common bread mould is

- a. *Penicillium*
- b. *Agaricus*
- c. *Aspergillus*
- d. **Rhizopus** **Ans**

29 0

30 The structure that absorbs nutrients from digested organic matter in fungi is

- a. Sporangiophore
- b. Sporangium
- c. **Rhizoid** **Ans**
- d. Spore

31 The most specialized protozoan in the following is

- a. *Paramecium* **Ans**
- b. *Euglena*
- c. *Amoeba*
- d. Trypanosome

32 Lung books or gill books are used for gaseous exchange in

- a. Crabs
- b. Termite
- c. Millipede
- d. **Spider** **Ans**

33 The subunits of lactose are

- a. Glucose + glucose
- b. **Glucose + galactose** **Ans**
- c. Glucose + fructose
- d. Fructose + fructose

34 What happen when red blood cells are placed in water

- a. the cells are non-affected
- b. the cells shrink and crinkles
- c. **the cells swell up and burst** Ans
- d. none

35 The intermediate product in anaerobic breakdown of glucose in fungi and germinating seeds is

- a. Citric acid
- b. Lactic acid
- c. **Pyruvic acid** Ans
- d. Glycolic acid

36 A stage of mitosis during which sister chromatids are pulled apart to opposite ends of the cell is

- a. Interphase
- b. Prophase
- c. Metaphase
- d. **Anaphase**Ans

37 A type of tactic response that involves the swimming of bacteria from cold regions to warm regions is

- a. Phototaxis
- b. Chemotaxis
- c. **Thermotaxis** Ans
- d. None

38 Example of joint that occurs at the elbow and knee is

- a. Ball and socket
- b. **Hinge** Ans
- c. Pivot
- d. Gliding

39 Which of the following has the largest surface area to volume ratio?

- a. Insect
- b. Man
- c. Hydra
- d. **Amoeba** Ans

40 The left and right parts of the heart are separated by

- a. **Septum** Ans
- b. Pericardium
- c. Ventricle
- d. Auricle

41 Why did AC woo PDP members? **Read the passages below and answer the questions that follow:**

Every discernible observer could foresee the crisis which engulfed the Edo House last week. Only a few days before, reports were rife about the AC reaching out to some PDP legislators to cross-carpet. The intention being to gain a majority and then assume the Speakership with the least constraint.

Zakawanu Garuba, the then Speaker, retorted with a fiat, threatening to declare vacant the seat of any defector. This in spite of the constitutional support for such action, as evidenced, for example, by the crisis that has torn the PDP apart in Edo State. The party has two factional chairmen in the state, even though one is more vociferous. The desperate attempt by Garuba to cling to the Speaker's chair is condemnable. He was not being recalled from the legislature; and so, he still has his seat as a floor member. As speaker, he was only first among equals. The Speakership is not his birthright. With the defection of one PDP lawmaker to the AC, the legislature reconvened hours after the bloodbath and elected a protem Speaker, while impeaching and suspending Garuba and a few others. They are to be probed. (Adapted from The Guardian, Thursday, March 4, 2011, p.14)

- a. They are few in number
- b. **They want speakership without stress** Ans
- c. PDP members are faithful
- d. The House is tough

42 The legislators impeached

- a. **Garuba the lawmakers** Ans
- b. PDP lawmakers
- c. Garuba and some PDP lawmakers
- d. Garuba and some lawmakers

43 The crisis in the Edo State House of Assembly could be predicted by

- a. The lawmakers
- b. Edo State indigenes
- c. The protem Speaker
- d. **All conscious observers** Ans

44 When I leave this country it will be for good. This means that I will **Choose the interpretation that you consider most appropriate for this sentences:**

- a. **never come back** Ans
- b. leave for better conditions elsewhere
- c. become an adventurer
- d. improvement ways

45 The committee has submitted its report the students. **Choose the word or group of words that best completes the following sentence:**

- a. in
- b. for
- c. about
- d. **on** Ans

46 The chairman told members that he was open suggestions. **Choose the word or group of words that best completes the following sentence:**

- a. to **Ans**
- b. for
- c. about
- d. on

47 Nobody will endure such profligate spending habits. **Choose the option that is nearest in meaning to the word underlined:**

- a. generous
- b. arrogant
- c. **wasteful** **Ans**
- d. mindless.

48 Is he oblivious of the political situation in his community? **Choose the option that is nearest in meaning to the word underlined:**

- a. **unconscious** **Ans**
- b. conscious
- c. afraid
- d. intolerant.

49 John and Ahmed's ideas were speculative **Choose the option that is opposite in meaning to the word underlined.**

- a. stipulative
- b. superlative
- c. **factual** **Ans**
- d. attractive.

50 Henry was admitted to the hospital with profuse bleeding. **Choose the option that is opposite in meaning to the word underlined.**

- a. much
- b. **little** **Ans**
- c. internal
- d. continuous

DAY 3

1 A spring scale reads 20 N as it pulls a 5 kg mass across a table. What is the magnitude of the force exerted by the mass on the spring scale?

- a. 49 N
- b. **20 N** **Ans**
- c. 5 N
- d. 4 N

2 A block slides down an inclined plane at angle θ to the horizontal. If the angle is increased, the coefficient of kinetic friction between the bottom surface of the block and the surface of the incline plane will

- a. Decrease
- b. Increase
- c. Generate heat
- d. Remain the same Ans**

3 Myopia is due to:

- a. older age
- b. change in focal length
- c. shortening of the eyeball
- d. elongation of the eyeball Ans**

4 Tesla is a unit of:

- a. electric flux
- b. magnetic flux
- c. electric field
- d. magnetic field Ans**

5 Which of the following consists of only vector quantities?

- a. force, mass and energy
- b. power, mass and energy
- c. velocity, power and acceleration
- d. velocity, acceleration and force Ans**

6 A stone sinks in water but floats in mercury. Why is this so?

- a. The stone is heavier than water but lighter than mercury.
- b. The stone is heavier than mercury but lighter than water.
- c. The stone is denser than water but less dense than mercury. Ans**
- d. The stone is denser than mercury but less dense than water.

7 A ball is released from the bottom of a swimming pool. It rises and reaches a constant speed after some time. At this moment, the resultant force acting on the ball is

- a. zero. Ans**
- b. upwards and less than the ball's weight.
- c. upwards and greater than the ball's weight
- d. upwards and equal to the ball's weight

8 A body is moving in a circle at constant speed. Which of the following statements about the body is true?

- a. There is no acceleration.
- b. There is no force acting on it. Ans**
- c. There is force acting at a tangent to the circle.
- d. There is force acting towards the centre of the circle.

9 A high resistance voltmeter reads 3V when connected across the terminal of battery on open circuit. What will be the voltmeter reading when the battery supplied a current of 0.2A to a lamp of 13 Ω ?

- a. 2.6V Ans**
- b. 0.52V

- c. 13V
- d. 1.5V

10 What is the height of an object 30cm from a pin - hole camera? If the height and image distance are 10cm and 20cm, respectively.

- a. 20cm
- b. 45cm
- c. **15cm Ans**
- d. 70cm

11 A car initially at rest travelled with acceleration of 5ms^{-2} , if its K.E. after 4s is 100000J, what is the mass of the car?

- a. 2000kg
- b. 500kg Ans
- c. 1000kg
- d. 5000kg

12 What is the value of a capacitor in series with two other capacitors valued $5\mu\text{F}$ and $7\mu\text{F}$ that are in parallel if the equivalent capacitance of the circuit is $4\mu\text{F}$.

- a. $6\mu\text{F}$ Ans
- b. $4\mu\text{F}$
- c. $2\mu\text{F}$
- d. $12\mu\text{F}$

13 Which of the following properties of light makes it behaves as a particle

- a. **Momentum Ans**
- b. diffraction and interference
- c. interference only
- d. diffraction only

14 A chemical substance which absorb moisture but does not turn to solution is called----

- a. hydrated
- b. efflorescence
- c. **hygroscopic Ans**
- d. deliquescent

15 The relative number of a reacting substance in a well-balanced chemical equation is

- a. mole
- b. stoichiometry coefficient
- c. **stoichiometry Ans**
- d. mole ratio

16 Brass is an alloy consists of

- a. **zinc and copper Ans**
- b. tin and copper
- c. zinc and tin
- d. copper and aluminium

- 17 What is the volume of 0.2 mole of oxygen gas at standard temperature and pressure?
- 2.24dm³
 - 4.48dm³** **Ans**
 - 1.12dm³
 - 22.4dm³
- 18 Which of these molecules has the highest entropy: H₂ (g), CO₂ (g), H₂O (l), CH₄ (g)?
- CO₂ (g)
 - H₂O (l)
 - C. H₂ (g)** **Ans**
 - CH₄ (g)
- 19 Rate of chemical reaction is not influence by -----
- temperature
 - catalysts
 - surface area
 - none** **Ans**
- 20 Purity of solid organic compounds can be tested physically by its
- boiling point
 - melting point** **Ans**
 - distillation
 - evaporation
- 21 A property that makes carbon atom unique and central to organic compound is
- functional group
 - isomerization
 - catenation **Ans**
 - epimerization
- 22 All these are aromatic compound except.....
- benzene
 - toluene
 - cyclohexane** **Ans**
 - phenol
- 23 An anti-knocking agent frequently added to gasoline to improve its performance is ---
- tetramethyl lead
 - tetraethyl lead** **Ans**
 - triethyl lead
 - trimethyl lead
- 24 Which of the following reagent(s) is/are appropriate for identifying metallic radicals?
- Sodium hydroxide

- b. Ammonium hydroxide
- c. Hydrochloric acid
- d. A & B Ans**

25 All the following do not form precipitate with sodium hydroxide solution except

- a. K^+
- b. NH_4^+
- c. Zn^{2+} Ans**
- d. Na^+

26..... gives lilac flame colour in a non-luminous Bunsen flame

- a. Na
- b. K Ans**
- c. Ca
- d. Ba.

27 Bilharziasis is caused by

- a. *Plasmodium*
- b. *Escherichia coli*
- c. Schistosome Ans**
- d. Retrovirus

28 Regulation of excess water in *Paramecium* is achieved by

- a. Food vacuole
- b. Contractile vacuole Ans**
- c. Gullet
- d. Trichocyst

29 Animals having both male and female sex organs in the same organisms are termed

- a. Hermaphrodites Ans**
- b. Parthenogenic
- c. Asexual
- d. Unisexual

30 One of these is an annelid except

- a. Ascaris Ans**
- b. Earthworm
- c. Leech
- d. Ragworm

31 Soft moist skin which can be used for gaseous exchange is a characteristic feature of

- a. Sharks
- b. Frog Ans
- c. Snake
- d. Eagle

32 The structure in which eggs are deposited in cockroach and earthworm is

- a. Pouch
- b. Cocoon Ans**
- c. Cloaca
- d. Ovipositor

33 Goiter is often associated with the deficiency of

- a. Chlorine
- b. Calcium
- c. Copper
- d. Iodine Ans**

34 The process by which liquid materials are taken up into the cell is termed

- a. Pinocytosis Ans**
- b. Phagocytosis
- c. Endocytosis
- d. Exocytosis

35 One of these is a trace element

- a. Potassium
- b. Calcium
- c. Manganese Ans**
- d. Phosphorus

36 The simple organic form of protein is

- a. Simple sugars
- b. Amino acids Ans**
- c. Glycerol
- d. Fatty acids

37 One of the following animals has hydrostatic skeleton

- a. Snake
- b. Earthworm Ans**
- c. Spider
- d. Mouse

38 A solid piece of bone below the neural canal is

- a. Neural canal
- b. Neural spine
- c. Centrum Ans**
- d. Neural arc

39 Open circulatory system is found in

- a. Birds
- b. Protozoans

c. **Molluscs** Ans

d. Man

40 The blood vessel that carries deoxygenated blood to the lung is

a. Pulmonary vein

b. **Pulmonary artery** Ans

c. Inferior vena cava

d. Superior vena cava

41 *Bolus* in the passage means **Read the passages below and answer the questions that follow:** Days passed fast for Ntanya and Teresa. They could hardly notice the land getting brown all over Kachawanga again, since neither of them hardly noticed anymore the sun come up and go down. Ntanya would work very hard in the field hauling in the last harvest and Teresa would stay at home washing, cooking and waiting, waiting for him to come home. She would cook his meal with extra care and when there was little meat she would only give bits to the children and not even touch it herself but use only the smell to get her food down and reserve the whole chunk for her husband. When Ntanya would come home with sweat on his brows she would put the wooden tray on a stool for him and sit down by him watching him eat with great satisfaction: every *bolus* that went down Ntanya's throat would also go down her own throat. When Ntanya would insist on sharing the food with her she would always say she had eaten already no matter how hungry she might be. Ntanya would then eat and drink to his satisfaction and always rest a while in the sun after saying 'Thank you mother'. (Taken from Peter K. Palangyo's novel: *Dying in the sun*)

a. throat pill

b. liquid

c. **piece of meat** Ans

d. large morsel.

42 Ntanya and Teresa could hardly notice the land getting brown because

a. they were on their honeymoon

b. they had just got married

c. **they hardly notice the sun** Ans

d. the sun did not come up

43 Ntanya said 'Thank you mother' to

a. his mother

b. **his wife** Ans

c. his aunt

d. Teresa's mother

44 When she confronted her husband, he hung his head and didn't answer her questions. This means that he was **Choose the interpretation that you consider most appropriate for this sentences:**

a. defiant

b. **ashamed and embarrassed** Ans

c. caught unawares

d. angry with her

45 The policeman shot the escaping armed robber the leg **Choose the word or group of words that best completes the following sentence:**

- a. on **Ans**
- b. in
- c. at
- d. into

46 For the management, your remarks amount to a slap the face **Choose the word or group of words that best completes the following sentence:**

- a. on
- b. upon
- c. at
- d. in **Ans**

47 Many people look into the future with trepidation. **Choose the option that is nearest in meaning to the word underlined:**

- a. certainty
- b. uncertainty
- c. fear **Ans**
- d. faith.

48 Can you marry such a lanky man? **Choose the option that is nearest in meaning to the word underlined:**

- a. tall and thin **Ans**
- b. sturdy
- c. fat and short
- d. thin and short.

49 Many people regard him as prolific. **Choose the option that is opposite in meaning to the word underlined.**

- a. diplomatic
- b. productive
- c. unproductive **Ans**
- d. unacademic.

50 Our success is paramount in his mind. **Choose the option that is opposite in meaning to the word underlined.**

- a. unimportant **Ans**
- b. important
- c. certain
- d. uncertain.

2018/2019 POST-UTME QUESTIONS

DAY 1

PHYSICS

1. A net force of 10 N accelerates an object at 5 m/s^2 . What net force would be required to accelerate the same object at 1 m/s^2 ?
 - A 1 N
 - B 5 N
 - C 2 N ++
 - D 50 N
2. A $2 \times 10^3 \text{ kg}$ car travels at a constant speed of 12 m/s around a circular curve of radius 30 m. As the car goes around the curve, the centripetal force is directed
 - A Tangentially to the curve in the direction of motion
 - B Toward the center of the circular curve ++
 - C Away from the center of the circular curve
 - D Tangentially to the curve opposite the direction of motion
3. The image formed by a concave mirror is observed to be virtual, erect and enlarged. The position of the object should be:
 - A. between F and C
 - B. at C
 - C. beyond C
 - D. between P and F ++
4. Ohm's law is not obeyed by:
 - A. electrolytes
 - B. discharge tubes
 - C. vacuum tubes
 - D. all of these ++
5. Mercury is used as a thermometric liquid in the thermometers because:
 - A. it wets the glass tube
 - B. it is a good conductor of heat ++
 - C. it cannot be obtained in pure state
 - D. mercury is transparent and not bright
6. The following are false of nuclear radiations except
 - A. alpha particles can be deflected by both electric and magnetic fields ++
 - B. beta particles are deflected electric field only

- C. gamma rays can be deflected by both electric and magnetic fields
- D. beta particle has the mass of helium

7. A machine gun with a mass of 5kg fires a bullet at a speed of 100ms^{-1} . The recoil speed of the machine gun is 1ms^{-1} , what is the mass of the bullet?

- A. 5g
- B. 50g ++
- C. 10g
- D. 20g

8. What is the latent heat of fusion of ice if the heat required to change 10kg of ice at 0°C into water at the same temperature is 800J?

- A. 8kg^{-1}
- B. 10kg^{-1}
- C. 80Jkg^{-1} ++
- D. 800Jkg^{-1}

9. If the current in each of the three 5Ω resistors connected in parallel is 12A, what is the potential difference applied across the combination?

- A. 40V
- B. 120V
- C. 60V ++
- D. 24V

10. If the cost of operating for 24 hours a lamp requiring 1A on a 220V line is 26.4Naira, the charges rate is?

- A. 24.2 N per kWh
- B. 4.2 N per kWh
- C. 24.4 N per kWh
- D. 4.40N per kWh ++

11. A potential difference of 6V is used to produce a current of 5A through a heating coil, the time spent to produced a heat of 6000J is

- A. 200s ++
- B. 60s
- C. 240s
- D. 24s

12. The EMF of an electric cell with internal resistance of 2Ω is 3.5V. If a current of 0.5A flows when a resistor is connected across it. Find the resistance of the resistor.

- A. $50\ \Omega$
 - B. $2.5\ \Omega$
 - C. $5\ \Omega$ ++
 - D. $1\ \Omega$
13. A guitar string is 50cm long. The half value of the wavelength of its fundamental note is
- A. 75cm
 - B. 150cm
 - C. 50cm ++
 - D. 37.5cm

CHEMISTRY

14. The electronic configuration of the phosphide ion $[P^{3-}]$ is
- A. $[Ne] 3S^2 3P^3$
 - B. $[Ne] 3S^2 3P^1$
 - C. $[Ne] 3S^2$
 - D. $[Ne] 3S^2 3P^6$. ++
15. The oxidation state of Chromium in $[Cr(OH)_6]^{3-}$ is
- A. +3 ++
 - B. +2
 - C. -3
 - D. -2
16. The ground state electronic configuration of copper is
- A. $[Ar] 3d^9 4s^1$
 - B. $[Ar] 3d^9 4s^2$
 - C. $[Ar] 3d^{10} 4s^1$ ++
 - D. $[Ar] 3d^{10} 4s^2$
17. What is the mass of 0.5mole of sodiumtrioxocarbonate (IV)? $[Na=23, O = 16, C = 12, H = 1]$
- A. 5.3g
 - B. 53g ++
 - C. 10.6g

D. 106g

18. What is the total number of valence electrons in $(\text{CH}_2)_2\text{CO}$?

A. 24 ++

B. 12

C. 18

D. 6

19. The splitting of a nucleus into two smaller nuclei of similar mass is called -----

A. fusion

B. Fission ++

C. decay

D. emission

20. Given the volume of carbon (IV) oxide gas at STP to be 2.24dm^3 , calculate the mole of carbon (IV) oxide gas

A. 0.10 mole ++

B. 0.20 mole

C. 0.30 mole

D. 0.40 mole

21. ----- law states that the volume of a given sample of gas at constant pressure is directly proportional to its absolute temperature

A. Boyle's

B. Avogadro's

C. Charles's ++

D. Graham's

22. The followings are factors affecting discharge of ions during electrolysis except -----

A. charge of ion ++

B. concentration of ion

C. nature of electrode used

D. position of the ion in electrochemical series

23. The imaginary mixing of different atomic orbital is called.....

A. orbitalisation

B. atomisation

C. Hybridization ++

D. isomerisation

24. Amine derivatives of carboxylic acid is called

- A. amino acid
- B. amide ++
- C. ammonolysis
- D. carboxylate

25. A test carried out to give additional support or evidence for the presence of species is called.....

- A. cation test
- B. flame test
- C. confirmatory test ++
- D. anion test

26. All these salts are soluble in water except.....

- A. nitrate salt
- B. sodium salts
- C. potassium salts
- D. carbonate salts ++

BIOLOGY

27. Regulation of constant internal environment is

- A. Osmosis
- B. Homeostasis ++
- C. Secretion
- D. Phototaxis

28. The most abundant component of blood plasma is

- A. Water ++
- B. Protein
- C. Glucose
- D. Urea

29. A condition whereby a person sees far objects clearly but nearby ones appear blurred

- A. Myopia
- B. Hypermetropia ++
- C. Astigmatism

- D. Cataract
30. The part of the ear responsible for maintenance of balance is
- A. Cochlea
 - B. Stapes
 - C. Semicircular canals ++
 - D. Ossicles
31. The ability to control or regulate the amount of light that enters the eye is
- A. Regulation
 - B. Accommodation ++
 - C. Vision
 - D. Photoreception
32. The vegetarian habit of the tadpole is related to possession of
- A. A streamline body
 - B. A long coiled intestine ++
 - C. External gills
 - D. A large head and a sucker
33. The hormone that lowers blood sugar level is
- A. Insulin ++
 - B. Glucagon
 - C. Thyrosine
 - D. Adrenaline
34. One of these is not a kidney disease
- A. Dropsy
 - B. Nephritis
 - C. Cirrhosis ++
 - D. Diuresis
35. In which of the following animals does external fertilisation occur?
- A. Toad ++
 - B. Bird
 - C. Lizard
 - D. Snake
36. A fertilised ovary forms
- A. Seed

- B. Pericarp
- C. Ovule
- D. Fruit ++

37. An example of a drupe is

- A. Coconut ++
- B. Tomato
- C. Pineapple
- D. Orange

38. The plant hormone that induces ripening of fruits and retard the growth of axillary buds is

- A. Ethene ++
- B. Cytokinin
- C. Gibberellin
- D. Auxin

39. If Ondo State population is currently 5 million and the annual population growth rate is 2%, what will the state's population be in 2020?

- A. 5.1 million
- B. 5.2 million
- C. 5.3 million ++
- D. 5.4 million

40. Increase in population beyond a point where available resources in the habitat are not adequate to support all individuals in the population is

- A. Overcrowding ++
- B. Over population
- C. Succession
- D. Maximum capacity

ENGLISH

Choose the interpretation that you consider most appropriate for this sentences:

41. The police knew that the suspect was not being straight with them. This means that the suspect was

- A. not telling the truth ++
- B. not friendly
- C. afraid

D. timid

42. After his prison experience, Etim decided to go straight. This means that he decided to

A. holds his head up and walk with defiance

B. change his religion

C. live an honest life ++

D stop using drugs

43. The teacher spoke with his tongue in his cheek when he said that Ade was the best pupil. This means that the teacher

A. did not mean what he said ++

B. was disappointed

C. was angry

D. tried to encourage

44. She does not remember to anyone at the peak

A. to talk

B. to be talking

C. talking ++

D. to have talked

45. This portion is to be shared between.....

A. us and them ++

B. we and them

C. us and they

D. we and they

46 If the Jury's verdict is "not guilty", the defendant will be

A. acquitted ++

B. indicted

C. pardoned

D. forgiven

Choose the option that is nearest in meaning to the word underlined:

47 The doctor diagnosed the tumour as malignant

A. deadly ++

B. painful

- C. trouble
- D. minor

48. The election system we adopted was his brainchild.

- A. undoing factor
- B. invention ++
- C. power
- D. fabrication

49. In all, I consider the behaviour unpalatable.

- A. unacceptable
- B. offensive
- C. acceptable ++
- D. inoffensive.

50. Her action will certainly exacerbate the situation.

- A. ameliorate ++
- B. worsen
- C. aggravate
- D. clarify.

DAY 2

PHYSICS

1. Which of the following is used to measure the internal diameter of a pipe?

- A. metre rule
- B. micrometer
- C. vernier calipers ++
- D. measuring cylinder

2. The motion of the pendulum from A to B and back to A is known as ____ oscillation.

- A. Two
- B. One and half
- C. One++
- D. None of the above

3. A bicycle takes 10s to travel the first 60 m, and another 20s to travel a further 90 m. What is the average speed?

- A. 4.5 m/s
- B. 5.0 m/s ++
- C. 6.5 m/s

D. 7.2 m/s

4. A moving body resist a change in its state of motion because of its

- A. weight
- B. mass
- C. speed
- D. Inertia ++

5. The echo of a sound from 1750m depth of an ocean is heard 2.5s later. What is the speed

- A. 3500m
- B. 1750m
- C. 1400 ms^{-1} ++
- D. 2800m

6. Calculate the wavelength of a wave traveling with velocity 360 ms^{-1} if the period of the wave is 0.17s

- A. 6m
- B. 17m
- C. 3m
- D. 60m ++

7. Calculate the focal length of an object placed 30cm from a concave mirror if the linear magnification of the image produced is 1

- A. 22cm
- B. 30cm
- C. 15cm ++
- D. 2.5cm

8. Elements that emit spontaneous radiation are

- A. stable
- B. Unstable ++
- C. radiogram
- D. isobar

9. An electric heater of 50W is used to heat a metal block of mass 10kg. In 5 minutes, by what value will the temperature rise if the specific heat capacity of the metal is $150 \text{ J kg}^{-1} \text{ K}^{-1}$.

- A. 15°C
- B. 50°C
- C. 12°C
- D. 10°C ++

10. If the energy stored in a capacitor is 0.5J when a 10 V supply is connected to it, what is the value of the capacitor?

- A. 1mF

- B. 10 Mf ++
- C. 0.5 mF
- D. 50 mF

11. The size of the image formed by an object 1cm high placed on the axis 15cm from a converging lens is 2cm, the image distance from the lens is.

- A. 30cm ++
- B. 15cm
- C. 2.5cm
- D. 1.2cm

12. Which of the following is not one of the peaceful applications of nuclear energy

- A. electricity generation
- B. research
- C. hand grenade ++
- D. destruction of cancer

13. _____ is emitted in a radioactive substance without altering the proton number of the substance?

- A. proton
- B. neutron ++
- C. alpha
- D. beta

CHEMISTRY

14. An haloacid with the highest acidic strength is -----

- A. HI ++
- B. HBr
- C. HCl
- D. HF

15. All the followings are polar molecules except -----

- A. HCN
- B. CH₃Cl
- C. H₂O
- D. CO₂ ++

16. There are discontinuities in electronic configuration of ----- and-----among the first series of transition elements

- A. Sc & Zn
- B. Fe & Co
- C. Cr & Cu ++
- D. Ti & Mn

17. The term used to describe the process of isolating a metal from its ore is called-----

- A. purification

- B. dissociation
 - C. extraction ++
 - D. modification
18. Which of these elements belong to salt former group?
- A. sodium
 - B. sulphur
 - C. Iodine ++
 - D. oxygen
19. A given mass of nitrogen is 0.12dm^3 and $1.01 \times 10^5 \text{N/m}^2$. Find its pressure at the same temperature if its volume is changed to 0.24dm^3
- A. $0.505 \times 10^5 \text{N/m}^2$ ++
 - B. $2.02 \times 10^5 \text{N/m}^2$
 - C. $4.04 \times 10^5 \text{N/m}^2$
 - D. $1.01 \times 10^5 \text{N/m}^2$
20. -----law explain the chemical behaviour of gases among all other gas laws
- A. Boyle's
 - B. Charcle's
 - C. Gay Lusaac's ++
 - D. Avogadro's
21. Properties that depend on the number of solute molecules and not on their chemical identity are called ----- properties
- A. non-chemical
 - B. isoelectric
 - C. Colligative ++
 - D. chemisorption
22. A system of reaction with ΔG equals to zero is said to be -----
- A. non-spontaneous
 - B. spontaneous
 - C. at equilibrium ++
 - D. semi- spontaneous
23. The process of converting vegetable oil into margarines is called -----
- A. substitution
 - B. Hydrogenation ++
 - C. halogenation
 - D. carboxylation
24. The most prominent reaction that benzene easily undergo is -----
- A. addition reaction
 - B. nucleophilic substitution reaction
 - C. electrophilic substitution reaction ++
 - D. elimination reaction

25..... s associated with brown ring formation on addition of FeSO_4 solution and concentrated H_2SO_4

- A. nitrate salt ++
- B. sulphate salts
- C. sulphite salts
- D. carbonate salts

26. Which of these chloride salts is insoluble in water?

- A. NaCl
- B. CaCl_2
- C. AgCl ++
- D. KCl

BIOLOGY

27. Which of these is an inner ear

- 27 Pinna
- 27 Incus
- 27 Tympanic membrane
- 27 Cochlea ++

28. The chisel-like shape teeth used for biting off pieces of food is called

- A. Canines
- B. Molars
- C. Premolars
- D. Incisors ++

29. The enzyme secreted by the cells lining the wall of stomach and which helps to coagulate milk is

- A. Rennin ++
- B. Pepsin
- C. Ptyalin
- D. Trypsin

30. The type of association between rabbit and the bacteria in its gut that secrete cellulase that digests cellulose is

- A. Saprophytism
- B. Parasitism
- B. Mutualism ++
- C. Commensalism

31. The portion of the alimentary canal of a bird where food materials are temporarily stored is

- A. Gizzard
- B. Crop ++
- C. Rectum
- D. Stomach

32. One of these is a male reproductive part of a flower

- A. Carpel
 - B. Stamen ++
 - C. Pistil
 - D. Sepal
33. Axial placentation is found in
- A. Cowpea
 - B. Pawpaw
 - C. Sunflower
 - D. Tomato ++
34. One of these is not a feature of a wind pollinated flower
- A. Light pollen grains
 - B. Dull colouration
 - C. No scent or nectaries
 - D. Bright colouration ++
35. The larval form of a beetle is called
- A. Wiggler
 - B. Grub ++
 - C. Maggot
 - D. Caterpillar
36. One of these is not a social insect
- A. Ant
 - B. Termite
 - C. Cockroach ++
 - D. Honeybee
37. Territoriality is common in one of the following organisms
- A. Dogs
 - B. Agama lizard ++
 - C. Snake
 - D. Deer
38. One of these is a biotic component of the ecosystem
- A. Humidity
 - B. Temperature
 - C. Water current
 - D. Bacteria ++
39. Which instrument can be used to measure light penetration in aquatic ecosystem?
- A. Rain guage
 - B. Secchi disc ++
 - C. Barometer
 - D. Anemometer
40. Which of these is a primary producer in terrestrial ecosystem?
- A. Fungi

- B. Grass ++
- C. Cow
- D. Lion

ENGLISH

Choose the interpretation that you consider most appropriate for this sentences:

41. They did not know what to expect and so decided to play it by it by ear. This means that they decided to

- A. Listen attentively to ear
- B. act according to circumstances ++
- C. pretend to be deaf
- D. prepare for the worst

42. Ojo decided that the best course of action was to play along with them for the time being. This means that he decided to

- A. join them temporarily
- B. entertain them for a while
- C. pretend to agree with them ++
be playful with them

D.

43. The conference was attended by people from all walks of life. This means that the conference was attended by people

- A. of all races
- B. who travelled widely
- C. from all profession ++
- D. from different parts of the world

44. When all the items on the agenda had been discussed, the meeting was

- A. adjourned ++
- B. dismissed
- C. postponed
- D. dissolved

45. Because the new venture was many businessmen and women went into it.

- A. rich
- B. manageable
- C. lucrative ++
- D. satisfying

46. We save a of lions at the game reserve

- A. pride ++
- B. family
- C. band

D. group

Choose the option that is nearest in meaning to the word underlined:

47. Amina is the one who initiated the quarrel, not Nike

- A. started ++
- B. stopped
- C. settled
- D. fuelled

48. Why should I retract my statement when it is nothing but the truth?

- A. alter
- B. deny
- C. withdraw ++
- Condemn
- D

49. Is he really behaving oddly?

- A. abnormally
- B. normally
- C. evenly
- D. Properly ++

50. Her health deteriorated rapidly.

- A. worsened
- B. Improved ++
- C. stagnated
- D. declined.

ENGLISH LANGUAGE

Read the passage below carefully and answer the questions that follow.

One of the most potent elements in body language is eye behaviour. You shift your eyes, meet another person's gaze or fail to meet it - and produce an effect out of all proportion to the muscular effort you have made. When two people look searchingly into each other's eyes, emotions are heightened and the relationship tipped toward greater intimacy. In normal conversation, each eye contact lasts only about a second before one or both individuals look away. Because the longer meeting of the eyes is rare, it is weighted with significance when it happens and can generate a special kind of human-to-human awareness. Most of the time, a lingering look is interpreted as a sign of attraction and this should be scrupulously avoided except in appropriate circumstances. A young woman once complained, 'That man makes me so uncomfortable, half the time when I glance at him he's already looking at me - and he keeps right on looking.' *Proper street behaviour requires a balance of attention and intention.* You are supposed to look at a passer-by just enough to show that you are aware of his presence. If you look too little, you appear haughty or furtive; too much and you are inquisitive. Usually what happens is that people eye each other until they are about eight feet apart, at which point both cast down their eyes. Much of eye behaviour is so subtle that we react to it only on the intuitive level. This has been demonstrated in elaborate experiments. Subjects sit and talk in the psychologist's laboratory, innocent of the fact that their eye behaviour is being observed from behind a one-way vision screen. In one fairly typical experiment, subjects were induced to cheat while performing a task, then were interviewed and observed. It was found that those who had cheated met the interviewer's eyes less often than was normal, an indication that 'shifty eyes' can actually be a tip-off to an attempt to deceive. However, none of the 'facts' of eye behaviour are cut and dried, for there are variations between individuals. People use their eyes differently and spend different amounts of time looking at others. Besides, no pattern of eye behaviour is precisely predictable in any normal conversation. Adapted from McQuade (1969), *Thinking in Writing*, p. 167

1. The young woman in the passage was uncomfortable because
 - a. the man appeared to be showing interest in her by his prolonged eye contact
 - b. the man's short and sharp gazes would tip the relationship toward greater intimacy
 - c. strangers who made the man's type of eye contact were likely to be dangerous
 - d. she was allergic to eye contact.

2. The reason given for the non-finality of research results on eye contact is
 - a. lack of concord and foresight among psychologists who carry out research on eye movement
 - b. lack of uniformity in eye behaviour and the variability of contact situation
 - c. that some subjects cheat during experiments, thus invalidating research findings
 - d. that research findings on eye contact have not been subjected to further empirical tests.

3. From the expression *proper street behaviour requires a balance of attention and intention*, it can be concluded that

- a. **even among strangers, attitudes and purposes can be deduced from eye behaviour**
- b. before walking the streets one must balance one's shifty eyes
- c. when intimate groups meet in the streets, eye contact is usually longer
- d. there is a balance between those with longer eye contact and those with shorter eye contact.

4. Which of the following can be concluded from the opening paragraph?

- a. The effects of eye contact are always overwhelming.
- b. **Eye behaviour is one of the non-verbal ways of communicating.**
- c. Eye movement is the most potent means of expressing intimacy.
- d. People are always offended by searching eye contact.

5. From the findings of the research described in the passage, one can reason that

- a. innocent people maintain longer eye contact than the guilty ones
- b. guilty people make less frequent gazes during interrogation
- c. 'shifty eyes' are equally induced by innocence and guilt
- d. **a feeling of guilt is occasioned by 'shifty eyes'.**

In each of following question, choose the interpretation that you consider *most appropriate* for these expressions.

6. The crisis ended as suddenly as it began.

- a. The crisis had suddenly begun.
- b. The crisis will not end suddenly.
- c. **The crisis stopped almost immediately.**
- d. The crisis will stop immediately.

7. One thing I will not be complaining about in my new job is a lack of excitement.

- a. The job is bad.
- b. **The job is exciting.**
- c. The job is not too exciting.
- d. The job has been previously done.

8. The Governor parried all the questions put to him by the journalist.

- a. The Governor answered all the questions brilliantly.
- b. **The Governor evaded all the questions.**
- c. The Governor failed all the questions.
- d. The Governor mastered all the questions.

9. If I visited England, I might go to Manchester City.
- a. When I go to England, I could go to Manchester City.
 - b. Whenever I visit England, I must go to Manchester City.
 - c. **I did not go to England and could not go to Manchester City.**
 - d. I could not visit Manchester City because I did not want to go to England.
10. The man puts his foot down whenever he is convinced of his action.
- a. He desires to assert his will in the situation.
 - b. He makes his mark wherever he goes.
 - c. His attitude demonstrates someone who likes to oppress others around him.
 - d. **He demonstrates firmness of character.**

In each question below, choose the option *nearest in meaning* to the word or phrase in *Italics*.

11. *Accountability* is certainly a desirable quality in a politician.
- a. Respectability
 - b. Courage
 - c. **Responsibility**
 - d. Diligence
12. The *common* practice among some media practitioners is to be sensational in their reporting.
- a. prevalent
 - b. **rampant**
 - c. ordinary
 - d. cogent
13. That *fateful* decision changed the company's outlook in many ways.
- a. wonderful
 - b. **disastrous**
 - c. uncontrollable
 - d. unsuccessful
14. The accident victim received *a superficial* wound from the crash.
- a. a serious
 - b. a painless
 - c. an internal
 - d. an external

BIOLOGY

1. In a dicot leaf, guard cells differ from other epidermal cells because they
 - a. have no definite shape
 - b. lack nuclei
 - c. are smaller
 - d. **contain chloroplasts**
2. Which of the following structures is NOT found in the female agama lizard?
 - a. Pre-anal pads
 - b. **Nuchal crest**
 - c. Gular fold
 - d. Nasal scale
3. If an isolated living cell is left in distilled water for two hours, it is likely to
 - a. lose some of its water to the surrounding water
 - b. lose all of its water to the surrounding water
 - c. reproduce by binary fission
 - d. **become more turgid.**
4. If an organic compound has its Hydrogen: Oxygen ratio as 2:1, it is likely to be
 - a. Water
 - b. **a carbohydrate**
 - c. a fat
 - d. a fatty acid and glycerol
5. Spirogyra and Mucor can be grouped together as Thallophyta because
 - a. they are unicellular organism
 - b. their spores could be dispersed by wind
 - c. they are capable of living independent lives
 - d. **their bodies are made up of thallus and filaments**
6. The mammalian organ through which nourishment and oxygen diffuse into a developing embryo is called a. **Placenta**
 - b. chorion
 - c. umbilical cord
 - d. oviduct
7. Which of the following organelles are likely to be present in cells that are actively respiring and photosynthesizing?
 - a. Nucleolus and centrioles
 - b. **Mitochondria and chloroplast**
 - c. Lysosomes and ribosomes
 - d. Golgi apparatus and endoplasmic reticulum
8. One of these organisms is not an Autotroph

- a. Spirogyra
 - b. Zea mays
 - c. Rhizobium
 - d. **Mushroom**
9. Muscles fatigue is caused by
- a. Accumulation of ethanol in the muscle cells
 - b. Accumulation of methanol in the muscles cells
 - c. **Accumulation of lactic acid in the muscle cells**
 - d. Accumulation of pyruvic acid in the muscle
10. Which of the organelles is not directly connected to cell division process?
- a. Centromere
 - b. Microtubules
 - c. **Golgi bodies**
 - d. Spindle apparatus
11. The mechanisms of opening and closing the stomata is associated with the
- a. **Guard cells**
 - b. Stoma
 - c. Lenticels
 - d. Air spaces
12. If a human skin cell with 46 chromosomes divided by mitosis, how many chromosomes will each daughter cell have?
- a. 23
 - b. 12
 - c. **46**
 - d. 92

CHEMISTRY

1. Basic hydrolysis of esters is called _____.
 a. acetylation
 b. acidification
 c. decarboxylation
 d. **saponification**
2. Carbon-carbon double bonds consist of _____ and _____.
 a. **1 sigma (δ) bond and 1 pi (π) bond**
 b. 2 sigma (δ) bond and 1 pi (π) bond
 c. 1 sigma (δ) bond and 2 pi (π) bond
 d. 2 sigma (δ) bond and 2 pi (π) bond
3. In stable organic compounds, carbon will always form _____ bond.
 a. 2

- b. **4**
 - c. 3
 - d. 5
4. What is the bond angle between the hybrid orbitals in methane?
- a. 180°
 - b. 120°
 - c. **109.5°**
 - d. 115.5°
5. Calculate the molar concentration of HCl if 50 ml of HCl is required to neutralise 25 ml of 1 M NaOH in acid base titration.
- a. 0.25 M
 - b. **0.5 M**
 - c. 1.0 M
 - d. 2.0 M
6. Which of the following is a bleaching agent?
- a. **sodium hypochlorite**
 - b. sulphur dioxide
 - c. liquid ammonia
 - d. ether
7. The aluminium belongs to group _____ and period _____ in the period table.
- a. 5 and 4
 - b. 5 and 2
 - c. **3 and 3**
 - d. 7 and 0
8. Aluminium is extracted from _____ by _____.
- a. **alumina, electrolysis**
 - b. alum, electrolysis
 - c. alumina, pyrolysis
 - d. alum, photolysis
9. The two major types of covalent bonds in carbonyl group are _____ and _____.
- a. **Sigma (δ) and pi (π) bonds**
 - b. pi (π) bonds and alpha (α)
 - c. Sigma (δ) and gamma (γ)
 - d. alpha (α) and beta (β)
10. The catalyst used in esterification reaction is _____.
- a. concentrated $C_6H_{12}O_6$
 - b. concentrated HN_3
 - c. concentrated CH_3Cl
 - d. **concentrated H_2SO_4**

11. Carbon dioxide contains carbon and oxygen in a fixed ratio 3:8 by mass. Name the law of chemical combination which governs this statement. **a. law of constant proportion**
 - b. Avogadro's Law
 - c. Boyle's law
 - d. Gay-Lussac Law
12. Ethanoic acid is a weak acid because?
 - a. it totally dissociates into its constituent ions when dissolved in water
 - b. it does not dissociate into its constituent ions when dissolved in water
 - c. **it only partially dissociates into its constituent ions when dissolved in water**
 - d. it completely dissociates into its constituent ions when dissolved in water

PHYSICS

1. Which of the following physical quantities is NOT a vector?
 - a. Velocity
 - b. **Work**
 - c. Force
 - d. Electric field intensity
 - e. Magnetic induction.
2. Of the following derived units, the one that is not a unit of power is
 - a. Joule/second
 - b. **Ampere volt**
 - c. Amphere²volt
 - d. Ohm²/volt
 - e. Volts²/ohm.
3. For which of the under listed quantities is the derived unit ML^2T^{-2} correct? I. Moment of a force II. Work III. Acceleration
 - a. I only
 - b. II only
 - c. **I and II**
 - d. II and III
4. Which of the following statements are true about the spring balance and the chemical balance? I. Both are used to measure the mass of an object. II. Either of them may be used to measure the weight of an object. III. The spring balance principle of moments. IV. A change in gravity changes the readings of a spring balance but not that of a chemical balance.
 - a. I and IV
 - b. II and III
 - c. **III and IV**
 - d. I, II, and III

- e. I and III.
5. Which of the following units is equivalent to kgms^{-1} ?
- Ns^{-1}
 - Nms
 - Ns**
 - Js^{-1}
 - sN
6. A football is hit with a velocity of 6ms^{-1} at an angle of 30° to the horizontal. The time of flight of the ball is
- 0.6s
 - 0.4s
 - 5s
 - 0.3s**
 - 3s
7. A machine gun with a mass of 5kg fires a 50g bullet at a speed of 100 ms^{-1} . The recoil speed of the machine gun is a.
- 0.5 m/sec
 - 1.5 m/sec
 - 1 m/sec**
 - 2 m/sec
 - 4m/sec.
8. A man stands on a spring scale placed in a lift. The lift descends at constant velocity. As a result, the scale reads a weight
- greater than the weight of the man
 - the same as the weight of the man**
 - of zero
 - less than the weight of the man
 - greater than the weight of the man by 1kg.
9. A rocket burns 0.01kg of fuel each second and ejects it as a gas with a velocity of 5,000m/s. What force does the gas exert on the rocket? a.
- 500 000N
 - 500N
 - 50N**
 - 5 000N
 - 50 000N
10. A 1000kg elevator is descending vertically with an acceleration of 1.0ms^{-2} , if the acceleration due to gravity is 10.0ms^{-2} , the tension in the suspending cable is
- 1.0N
 - 10.0N
 - 9 000.0N**
 - 11 000.0N

e. 9.0 N

11. When taking a penalty kick, a footballer applies a force of 30.0N for a period of 0.05s. If the mass of the ball is 0.075kg, calculate the speed with which the ball moves off. a.

4.50ms⁻¹

b. 11.25ms⁻¹

c. **20.00ms⁻¹**

d. 45.00ms⁻¹

e. 21 ms⁻¹

12. If it takes 5.0 hours to drain a container of 540.0cm³ of water, what is the flow rate of water from the container in kgs⁻¹? (Density of water 1000 kgm⁻³) a. 32.5

b. 31.5

c. 30.8

d. **30.0**

e. 22

MATHEMATICS

1. A group of market women sell at least one of yam, plantain and maize. 12 of them sell maize, 10 sell yam and 14 sell plantain. 5 sell plantain and maize, 4 sell yam and maize, 2 sell yam and plantain only while 3 sell all the three items. How many women are in the group?

a. **25**

b. 19

c. 18

d. 17

2. Find all real number x which satisfy the inequality $\frac{1}{3}(x + 1) - 1 > \frac{1}{5}(x + 4)$

a. $x < 11$

b. $x < -1$

c. $x > 6$

d. $x > 11$

0.02174 $\times 1.2047$

3. Express each number in to two significant figures and then 0.023789 evaluate.

a. 0

- b. 0.9
- c. 1.1
- d. 1.2
4. Four boys and ten girls can cut a field first in 5 hours. If the boys work at $\frac{5}{4}$ the rate at which the girls work, how many boys will be needed to cut the field in 3 hours?
- a. 180
- b. 60
- c. 25
- c. 20
5. Find the value of X if $\frac{\sqrt{2}}{x+\sqrt{2}} = \frac{1}{x-\sqrt{2}}$
- a. $3\sqrt{2+4}$
- b. $3\sqrt{2-4}$
- c. $3-2\sqrt{2}$
- d. $4+2\sqrt{2}$
6. A cylindrical tank has a capacity of 3080 m³. What is the depth of the tank if the diameter of its base is 14 m?
- a. 25 m
- b. 23 m
- c. 22 m
- d. **20 m**
7. QST The acres for rice, pineapple, cassava, cocoa and palm oil in a certain district are given respectively as 2, 5, 3, 11 and 9. What is the angle of the sector of cassava in a pie chart?
- a. 180°
- b. **36°**
- c. 60°
- d. 108°
8. Three consecutive terms of a geometric progression are given as $n-2$, n and $n+3$. Find the common ratio. a. $\frac{3}{2}$
- b. $\frac{4}{3}$
- c. $\frac{1}{2}$

- d. $\frac{1}{4}$
9. In a class of 40 students, 32 offer mathematics, 24 offer physics and 4 offer neither mathematics nor physics. How many offer both mathematics and physics?
- a. 4
 - b. 8
 - c. 16
 - d. **20**
10. The sum of the interior angle of pentagon is $6x + 6y$. Find y in terms of x .
- a. $y=6-x$
 - b. $y=90-x$
 - c. $y=120-x$
 - d. $y=150-x$
11. How many sides has a regular polygon whose interior angle is 135°
- a. 12
 - b. 10
 - c. 9
 - d. **8**
12. The locus of a dog tethered to a pole with a rope of 4 m is a
- a. circle with diameter 4 m
 - b. **circle with radius 4 m**
 - c. semi-circle with diameter 4 m
 - d. semi-circle with radius 4 m

English

Choose the word or phrase that is *most nearly opposite* in meaning to the italicized word and that will at the same time, correctly fill the gap in the sentence

1. The management has neither _____ nor *denied* the rumour that is was going to lay off some workers.
- a. pronounced
 - b. ****confirmed**
 - c. rejected
 - d. advertised

2. Instead of _____ the issue, the speaker *digressed* to unimportant matters.

- a. summarizing
- b. describing
- c. revealing
- d. **addressing

3. The telephone booths were all _____ and I had to wait for more than ten minutes for one to become *vacant*. a. locked

- b. busy
- c. **occupied
- d. ringing

4. While the first battalion *halted* its march at nightfall, the second _____ to the barracks

- a. returned
- b. set out
- c. **proceeded
- d. turned up

Choose the words that *best complete* the following sentences

5. If the jury's verdict is "not guilty," the defendant will be _____

- a. **acquitted
- b. indicted
- c. paroned
- d. forgiven

6. When all the items on the agenda had been discussed, the meeting was _____

- a. **adjourned
- b. dismissed
- c. postponed
- d. dissolved

7. Because the new venture was _____ many businessmen and women went into it.

- a. rich
- b. manageable
- c. **lucrative
- d. satisfying

8. We save a _____ of lions at the game reserve

- a. **pride

- b. family
- c. band
- d. group

9. Many educationists feel that _____ Punishment should still be retained in schools

- a. capital
- b. **corporal
- c. corporate
- d. physical

Choose from the options the word that has the same vowel sound in italics as the one represented by the letter(s)

10. *Rust*

- a. loud
- b. touch
- c. bought
- d. **march

11. Res*ear*ch

- a. disturb
- b. comfort
- c. **affair
- d. carry

12. B*oa*t

- a. board
- b. bought
- c. **go
- d. glory

13. *Air*

- a. **heir
- b. her
- c. hear
- d. fire

14. Ple*a*sure

- a. pliable
- b. **pleat
- c. plenary

d. jeopardy

BIOLOGY

1. The hypha of rhizopus is said to be non-septate because it
 - a. Does not contain chlorophyll
 - b. **Has no cross wall
 - c. Is vacuolated
 - d. Stores oil globus
2. The butterfly is of great economic importance because
 - a. Of its use in scientific studies
 - b. It sucks nectar from flowers
 - c. It adds to the beauty of the environment
 - d. **It pollinates flowers of crops and other plants
3. In which of the following does external fertilization take place?
 - a. **Toad
 - b. Lizard
 - c. Bird
 - d. Cockroach
4. Growing yam tendrils climb for support. This growth response is
 - a. **Haptotropism
 - b. Geotropism
 - c. Phototropism
 - d. Hydrotropism
5. The bright colours of the comb and feathers in the peacock are for?
 - a. Sex differentiation
 - b. Beauty
 - c. **Courtship
 - d. Defence
6. An evidence of a common ancestral for fishes, amphibians, reptiles, birds and mammals is the
 - a. Possession of wings by birds and bats
 - b. Cold-bloodedness of fishes, amphibians and reptile
 - c. **Presence of gill clefts in vertebrate embryos
 - d. Possession of scale by fish and reptiles
7. Which of the following is a homostatic response in humans?
 - a. Withdrawing the hand from a hot object
 - b. The mouth getting watery when food is sighted
 - c. Yawning owing to tiredness

- d. ****Shivering in a cold environment**
8. A man with a normal haemoglobin (AA) marries a woman who has sickle cell haemoglobin (SS). They have a child who has sickle-cell trait. Which of the following genotypes could be associated with the child's haemoglobin?
- a. AA
- b. ****AS**
- c. AO
- d. SS
9. A test tube containing yeast in glucose solution containing alkaline pyrogallol.
- The bubbles of carbon (IV) oxide produced indicates that the yeast cells are
- a. ****Respiring in the absence of oxygen**
- b. Liberating oxygen on their own
- c. Living and consuming oxygen
- d. Being killed by the alcohol produced
10. The glottis is the opening which leads to the
- a. Oesophagus
- b. ****Larynx**
- c. Nostrils
- d. Pharynx
11. The carrier of the hereditary characters in plants is the
- a. ****Gene**
- b. Nucleus
- c. Chromosome
- d. Chloroplast
12. Which of the following sources is not an evidence of evolution?
- a. Fossil records
- b. Comparative anatomy
- c. ****Human behavior**
- d. Comparative embryology

CHEMISTRY

1. Faraday's constant is _____ Coulomb mole⁻¹
- a. ****96485**
- b. 96500
- c. 650
- d. 965
2. Galvanic cell is _____.

- a. rechargeable
 - b. lead battery
 - c. **none-rechargeable
 - d. all of the above
3. Sulphuric acid (H_2SO_4) is a _____.
- a. **strong electrolyte
 - b. non-electrolyte
 - c. weak electrolyte
 - d. none of the above
4. Which one of the following will turn red litmus blue?
- a. Vinegar
 - b. **Baking soda solution
 - c. Lemon juice
 - d. Soft drinks
5. Methyl orange is _____.
- a. **Pink in acidic medium, yellow in basic medium
 - b. Yellow in acidic medium, pink in basic medium
 - c. Colourless in acidic medium, pink in basic medium
 - d. Pink in acidic medium, colourless in basic medium.
6. Lime water is _____.
- a. CaO
 - b. ** $\text{Ca}(\text{OH})_2$
 - c. CaCO_3
 - d. CaCl_2
7. The summation of pH and pOH is equal to _____
- a. 10
 - b. 7
 - c. **14
 - d. 12
8. The chemical formula of caustic potash is _____.
- a. NaOH
 - b. $\text{Ca}(\text{OH})_2$
 - c. NH_4OH
 - d. ** KOH
9. Isotopes of an element have a different number of

- a. Proton
- b. **Neutron
- c. Electron
- d. atom

10. What is the product formed at the cathode in the electrolysis of aqueous CuSO_4 ?

- a. **Copper metal
- b. Oxygen gas
- c. Hydrogen gas
- d. Sulphur

11. Pressure affects the rate of reaction involving -----

- a. solids
- b. liquids
- c. *gases
- d. semi liquid

12. When the rate of forward and backward reactions is equal, the reaction is said to be in_____.

- a. **equilibrium
- b. catalytic reaction
- c. heterogenous reaction
- d. homogenous reaction

QST Which of the following is a noble gas?

- a. **He
- b. S
- c. F
- d. Al

PHYSICS

1. A man 1.5m tall is standing 3m in front of a pinhole camera whose distance between the hole and the screen is 0.1m. What is the height of the image of the man on the screen. a. **0.05m

- b. 0.15m
- c. 0.30m
- d. 1.00m
- e. 0.5m.

2. Convex mirrors are used as driving mirrors because images formed are

- a. **erect, virtual and diminished
- b. erect, real and diminished
- c. erect, virtual and magnified

- d. inverted, virtual and diminished
- e. enlarged.

3. The image of an object is located 6cm behind a convex mirror. If its magnification is 0.6, calculate the focal length of the mirror a. 3.75cm

- b. 6.60cm
- c. 10.00cm
- d. **15.00cm
- e. 30.0 cm.

4. When a ray of light passes from the water to glass, it is

- a. bent away from the normal
- b. **bent towards the normal
- c. totally internally reflected
- d. absorbed by the glass
- e. broken into pure spectrum.

5. A 45° triangular glass prism can be used as a reflector of light because

- a. refraction never takes place in such a prism
- b. the angle of reflection equals the angle of incidence
- c. the refractive index of glass is less than 1
- d. it is transparent
- e. **the critical angle for glass is less than 45° .

6. A submarine is observed to rise from a real depth of 80m to 60m in water. Calculate the change in apparent depth. (refractive index of water = $\frac{4}{3}$).

- a. 60m
- b. **15m
- c. 80m
- d. 45m
- e. 20m

7. The lack of power of accommodation which is mainly due to the hardening of the eye lens and is usual with the onset of old age is called

- a. myopia
- b. hypermetropia
- c. **presbyopia
- d. astigmatism
- e. eye ring.

8. A certain far-sighted person cannot see objects that are closer to the eye than 50cm clearly. Determine the power of the converging lens which will enable him to see at 25cm. a. 0.03D

- b. 0.04D
- c. 0.06D
- d. **0.02D

e. 0.032 D.

9. Which of the following is a secondary colour?

- a. blue
- b. **orange
- c. red
- d. green

10. Which of the following electromagnetic waves has the highest frequency a. infrared rays

- b. **X-rays
- c. Ultra-violet rays
- d. radio waves
- e. microwaves.

11. The optical phenomenon responsible for rainbow is

- a. refraction
- b. diffusion
- c. reflection
- d. **Dispersion
- e. interference.

12. Why is it impossible for the lines of force of an electric field to cross one another

- a. like charges repel each other
- b. a charged particle is deflected
- c. like changes are arranged on the same lines of force
- d. **a positively charged particle can travel in only one direction at any time
- e. none of these is correct.

MATHEMATICS

1. A cylindrical tank has a capacity of 3080 m^3 . What is the depth of the tank if the diameter of its base is 14m? Take $\pi = 22/7$.

- a. 23 m
- b. 25 m
- c. **20 m
- d. 22 m

2. The value of $x + x (x^x)$ when $x = 2$ is

- a. 16
- b. **10
- c. 18
- d. 24
3. A man covered a distance of 50 miles on his first trip, on a later trip he traveled 300 miles while going 3 times as fast. His new time compared with the old distance was?
- a. three times as much
- b. the same
- c. **twice as much
- d. half as much
4. Divide $4x^3 - 3x + 1$ by $2x - 1$
- a. $2x^2 - x + 1$
- b. ** $2x^2 - x - 1$
- c. $2x^2 + x + 1$
- d. $2x^2 + x - 1$
5. A car dealer bought a second-hand car for of 250,000 and spent N 70,000 refurbishing it. He then sold the car for N400,000. What is the percentage gain?
- a. 60%
- b. 32%
- c. **25%
- d. 20%
6. Find the number of ways that the letters of the word EXCELLENCE be arranged
- a. $\frac{10!}{2!2!2!}$
- b. ** $\frac{10!}{4!2!}$
- c. $\frac{10!}{4!2!2!}$

d. $\frac{10!}{2!2!}$

7. If a rod 10cm in length was measured as 10.5cm, calculate the percentage error

- a. **5%
- b. 10%
- c. 8%
- d. 7%

8. Find the principal which amounts to ₦ 5,500 at a simple interest in 5 years at 2% per annum

- a. #4,900
- b. **#5000
- c. #4700
- d. #4000

9. y is inversely proportional to x and y and 6 when x = 7. Find the constant of the variation

- a. 47
- b. **42
- c. 54
- c. 46

10. Find the equation of the locus of a point p (x, y) such that $p_v = p_w$, where $v = (1, 1)$ and $w = (3, 5)$

- a. $2x + 2y = 9$
- b. $2x + 3y = 8$
- c. $2x + y = 9$
- d. ** $x + 2y = 8$

11. If $m * n = [m_n - n_m]$ for m, n belong to R, evaluate $-3 * 4$

- a. 3
- b. 4
- c. **5
- d. 6

12. In how many ways can the word MATHEMATICS be arranged?

- a. $\frac{11!}{9!2!}$
- b. $\frac{11!}{9!2!2!}$
- c. $\frac{11!}{**2!2!2!}$
- d. $\frac{11!}{2!2!}$

ENGLISH

Choose the words that **best complete** the following sentences

1. Surprisingly, Musa turned _____ the job offer
 - a. away
 - b. in
 - c. out
 - d. **down
2. We have no choice _____ to do what they demand.
 - a. except
 - b. than
 - c. unless
 - d. **but
3. The committee has submitted its report _____ the students
 - a. in
 - b. for
 - c. about
 - d. **on
4. The chairman told members that he was open _____ suggestions

- a. **to
- b. for
- c. about
- d. on

5. The policeman shot the escaping robber _____ the leg

- a. **on
- b. in
- c. at
- d. into

6. For the management, your remarks amount to a slap _____ the face

- a. on
- b. upon
- c. at
- d. **in

7. QST She does not remember _____ to anyone at the park

- a. to talk
- b. to be talking
- c. **talking
- d. to have talked

Fill the gaps with the correct options:

8. University students _____ behave well.

- a. can
- b. **should
- c. may
- d. must

9. Gloria _____ have finished the project two days ago.

- a. **Must
- b. may
- c. can
- d. supposed to

10. We shall see you soon, _____?

- a. shall we?
- b. **shan't we?
- c. shouldn't we
- d. should we

11. He _____ not come now

- a. Needs
- b. needed
- c. **need
- d. will need

In each of following question, choose the interpretation that you consider *most appropriate* for these expressions.

12. They did not know what to expect and so decided to play it by it by ear. They decided to

- a. listen attentively to everyone
- b. **act according to circumstances
- c. pretend to be deaf
- d. prepare for the worst

13. Ojo decided that the best course of action was to play along with them for time being. He decided to

- a. join them temporarily
- b. entertain them for a while
- c. **pretend to agree with them
- d. be playful with them.

14. The conference was attended by people from all walks of life. The conference was attended by people a. of all races

- b. who travelled widely
- c. **from all professions
- d. from different parts of the world.

BIOLOGY

1. The most common characteristic of the fungal hyphae is the possession of

- a. **Cell-like compartment with one nucleus each
- b. Cell walls that are both rigid and chitinous
- c. A multicellular mycelium in the substrate
- d. Cell-like partitions formed by cross-walls

2. The larval stage of a mosquito is called

- a. **Wiggler
- b. Grub
- c. Maggot
- d. Caterpillar

3. Which of the following is not present in RNA?

- a. Adenine
- b. Guanine
- c. Cytosine
- d. **Thymine

4. Which of the following is an example of parasitism
- Fungi growing on a dead tree branch
 - A squirrel living in an abandoned nest of a bird
 - **Mistletoe growing on an orange tree**
 - Cattle egrets taking ticks from the body of cattle
5. A peculiar characteristics of mammals is that they
- Have teeth
 - Are warm-blooded
 - Have lungs
 - **Have sebaceous glands**
6. QST Alternation of asexual modes of reproduction is found in
- Blue-green algae
 - Euglena
 - **Fern**
 - Maize
7. The highest level of ecological organization is the
- Ecosystem
 - Population
 - **Biosphere**
 - Niche
8. Beriberi is caused by deficiency of vitamin
- **B**
 - C
 - A
 - E
9. QST The hormone which regulates the amount of glucose in the blood is called?
- Adrenalin
 - Auxin
 - **Insulin**
 - Thyroxine
10. Which of the following insects lays its eggs in horny, purse-like cases?
- Mosquito
 - Butterfly
 - Termite
 - **Cockroach**

11. The group of insects that have mouth parts adapted for both piercing and sucking is
- Cockroach, aphids and mosquitoes
 - Aphids, houseflies and moths
 - **Mosquitoes, tse-tse flies and aphids**
 - Aphids, beetles and grasshoppers
12. Which of the following does a virus have in common with animal cells?
- Nucleus
 - **DNA**
 - Gycogen
 - Cytoplasm

CHEMISTRY

1. Which of the following does not exist?
- **methene**
 - methanal
 - butanone
 - propanal
2. Reaction of methanal with NaBH_4 gives _____.
- methane
 - **methanol**
 - ethanol
 - propanone
3. Which substance is called universal solvent?
- hexane
 - **water**
 - ethanol
 - glucose
4. Which of the following are the correct allotropes of sulphur?
- **yellow rhombic sulphur (α -sulphur) and the monoclinic (β -sulphur)**
 - yellow rhombic sulphur (α -sulphur) and the tetraclinic (δ -sulphur).
 - green rhombic sulphur (γ -sulphur) and the monoclinic (β -sulphur).
 - black rhombic sulphur (μ -sulphur) and the monoclinic (β -sulphur).
5. Which of the following is not an isotope of hydrogen
- protium
 - deuterium
 - tritium

- d. **radium
6. The shape of water molecule is _____ shape.
- trigonal
 - linear
 - **V (bent)
 - trigonal planar
7. The type water which produces large amount of lather with soap is known as _____ water.
- ** soft
 - hard
 - amorphous
 - lime
8. The type of hardness that is due to the presence of chlorides and sulphates of calcium and magnesium is _____ hardness.
- temporary
 - **permanent
 - amorphous
 - gel
9. What are the products of this reaction?
 $\text{CaO(s)} + \text{HCl(g)} \rightarrow ? + ?$ a.
- $\text{CaHCO}_3, \text{H}_2\text{O}$
- $\text{CaCO}_3, \text{H}_2\text{O}$
 - $\text{H}_2\text{O}, \text{CaO}$
 - ** $\text{CaCl}_2, \text{H}_2\text{O}$
10. How many isomers are possible for Heptane (C_7H_{16}) a.
- 4
 - **9
 - 8
 - 7
11. Which chemical can be used in the laboratory to make oxygen?
- Hydrochloric Acid
 - **Hydrogen peroxide
 - Sodium hydroxide
 - All of the above
12. What catalyst is usually used in the laboratory to speed up oxygen production? a.
- Iron
- Vanadium (V) oxide
 - **Manganese (IV) oxide
 - None of the above

PHYSICS

1. A force, 10N, drags a mass 10kg on a horizontal table with an acceleration of 0.2ms^{-2} . If the acceleration due to gravity 10ms^{-2} , the coefficient of friction between the moving mass and the table is a. 0.02
 - b. 0.08
 - c. 0.20
 - d. **0.80
 - e. 2.

2. In which of the following is surface tension important?
 - a. The floating of a ship in water
 - b. **The floating of a dry needle in water
 - c. The floating of a balloon in air
 - d. The diffusion of a sugar solution across membrane.
 - e. Dynamics.

3. A piece of wire gauze can be made to lie on water because
 - a. the wire gauze is less dense than water
 - b. the water molecules repel the wire gauze upwards
 - c. **the water surface has the effect of an elastic thin skin
 - d. of the cohesive forces between the water and the wire gauze molecules
 - e. Elastic modulus.

4. When the brakes in a car are apply, the frictional force on the tyre is
 - a. A disadvantage because it is in the direction of motion of the car
 - b. A disadvantage because it is in the opposite direction of motion of the car
 - c. An advantage because it is direction of motion of the car
 - d. **An advantage because it is in the option motion of the car
 - e. Disadvantage because of heat.

5. Water does not drop through an open umbrella of silk material unless the inside of the umbrella is touched. This is due to a. capillarity
 - b. Osmotic
 - c. viscosity
 - d. **surface tension.
 - e. diffusion.

6. As the pressure of a fluid increases, its viscosity
 - a. Decreases
 - b. **Remains constant

- c. increases then decreases
 - d. increases
 - e. Absolute.
7. A 100kg box is pushed along a road with a force of 500N. If the box moves with a uniform velocity, the coefficient of friction between the box and the road is a. **0.5
- b. 0.4
 - c. 1.0
 - d. 0.8
 - e. 0.9
8. Counting of currency notes with moist fingers is based on the principles of
- a. diffusion.
 - b. cohesion.
 - c. **adhesion
 - d. viscosity.
 - e. Osmosis.
9. The terminal velocity of a ball-bearing falling through a viscous fluid is reached when the
- a. up thrust is equal to the weight of the ball.
 - b. ball accelerates uniformly.
 - c. up thrust is equal to the velocity of the ball.
 - d. **velocity is uniform.
 - e. upthrust is low.
10. The surfaces of conveyor belts are made rough so as to
- a. **prevent the load from slipping
 - b. make them stronger.
 - c. enable them to carry more load
 - d. protect them while carrying loads.
 - e. Carry less.
11. The friction which exists between two layers of liquid in relative motion is called a.
- capillary
 - b. Surface
 - c. tension
 - d. **viscosity
 - e. Cohesion.
12. Which of the following statements defines correctly the efficiency of a machine?
- a. The load carried by the machine divided by effort required in carrying the load
 - b. The distance moved by the load divided by the distance moved by the effort

- c. **The useful work done by the machine divided by the total work put into it.
- d. The movement of the load divided by the movement of the effort about the same point.
- e. The temperature at the end of the application of the machine divided by the initial temperature.

MATHEMATICS

1. If line $p = 5x + 3$ is parallel to line $p = wx + 5$. Find the value of w .

- a. 7
- b. 3
- c. 6
- d. **5

2. Evaluate

$$\int (\cos 4x + \sin 3x) dx$$

- a. $\sin 4x - \cos 3x + k$
- b. $\sin 4x + \cos 3x + k$
- c. $\frac{1}{4} \sin 4x - \frac{1}{3} \cos 3x + k$
- d. $\frac{1}{4} \sin 4x + \frac{1}{3} \cos 3x + k$

3. Calculate the range of 20, -6, 25, 30, 21, 28, 32, 33, 34, 5, 3, 2, and 1.

- a. 32
- b. 36
- c. 33
- d. **40

4. Calculate the perimeter of a sector of a circle of radius 12cm and angle 60° .

- a. $(12 + 4\pi)$ cm
- b. $^{**}(24 + 4\pi)$ cm
- c. $(12 + 6\pi)$ cm
- d. $(24 + 6\pi)$ cm

5. The table above shows the frequency distribution of marks obtained by a group of students. If the total mark is 48, find the value of y .

Marks	2	3	4
Frequency	4	4	y

- a. 6
- b. 8
- c. **7

- d. 5
6. Given $U = \{x \mid x \text{ is a positive integer less than } 15\}$ and $P = \{x \mid x \text{ is even number from } 1 \text{ to } 14\}$. Find the complement a. $\{1, 3, 5, 7, 9, 11, 13, 15\}$
 b. $\{2, 3, 5, 7, 9, 11, 13\}$
 c. $\{1, 3, 5, 7, 9, 11, 13\}$
 d. $\{2, 3, 5, 7, 9, 11, 15\}$
7. A number of pencils were shared out among Bisi, Sola and Tunde in the ratio of 2:3:5 respectively. If Bisi got 5, how many were shared out? a. 15
 b. 25
 c. 30
 d. 50
8. Evaluate $\int_0^{\pi/2} \sin x dx$
 a. 2
 b. -1
 c. 1
 d. -2
9. An arc of the length 16π cm subtends an angle of 80° at the centre of the circle. Find the radius of the circle.
 a. 24 cm
 b. 28 cm
 c. 36 cm
 d. 32 cm
10. The n th term of the sequence 3, 9, 27, 81..... is
 a. $3 \times 3^{n-2}$
 b. $3 \times 3^{n-1}$
 c. $3 \times 3^{n+2}$
 d. $3 \times 3^{n+1}$
11. Simplify $1 - (\frac{1}{7} \times 3 \frac{1}{2}) \div \frac{3}{4}$
 a. 2
 b. $\frac{1}{3}$
 c. 1
 d. $\frac{2}{3}$

12. QST If a car travels 120km on 45 litres of petrol, how much petrol is needed for a journey of 600km?
- a. 720 litres
 - b. 160 litres
 - c. **225 litres
 - d. 960 litres

UNIVERSITY OF MEDICAL SCIENCES, ONDO
POST UNIFIED TERTIARY MATRICULATION SCREENING
2023

English

Choose the option that is most nearly opposite in meaning to the underlined word and that will, at the same time, correctly fill the gap in the sentence

His father is an..... driver but he is only a novice

- a. Enlightened
- b. Excellent
- c. poor
- d. experienced.

The arrival of the police at the scene..... rather than mitigate tension

- a. Provoked
- b. **Ans** heightened
- c. created
- d. prolonged.

Roses are common in May but.... in November.

- a. dead
- b. strange
- c. **Ans** rare
- d. extinct.

A bank clerk should be able to tell easily the difference between.... Notes and counterfeit ones.

- a. crisp
- b. good
- c. correct
- d. **Ans** genuine.

Last night, Uncle Joe was blind drunk but this morning he is as..... as a judge

- a. Alert
- b. saintly
- c. strong
- d. **Ans** sober.

Choose the word or group of words that best completes the following sentence:

Surprisingly, Musa turned the job offer

- a. Away
- b. In
- c. out
- d. **Ans** down

We have no choice to do what they demanded.

- a. except
- b. **Ans** than
- c. unless
- d. but

The committee has submitted its report the students.

- a. in
- b. for
- c. about
- d. **Ans** on

The chairman told members that he was open suggestions

- a. **Ans** to
- b. for
- c. about
- d. on

The policeman shot the escaping armed robber The leg.

- a. on
- b. **Ans** in
- c. at
- d. into

For the management, your remarks amount to a slap the face

- a. **Ans** on
- b. upon
- c. at
- d. in

Biology

The presence of hooks and suckers in tapeworm is an adaptation to what mode of life

- a. Commensal
- b. **Ans** Parasitic
- c. Saprophytic
- d. Mutualistic

Plants that can survive in an environment where water is scarce are

- a. **Ans** Xerophytes
- b. Hydrophytes
- c. Mesophytes
- d. Thermophytes

A short thick beak adapted for crushing seeds is a feature of

- a. Sunbird
- b. **Ans** Sparrow
- c. Eagle
- d. Duck

A condition that leads to excessive algal growth as a result of pollution of aquatic ecosystem is

- a. Introphication
- b. Ultrafiltration
- c. **Ans** Eutrophication
- d. Hydrophication

A constituent of the exhaust gases of motor vehicles that causes serious air pollution is

- a. Water vapour
- b. Ozone
- c. **Ans** Carbon monoxide
- d. Carbon dioxide

Who proposed the theory of use and disuse of body parts in evolutionary theory

- a. Darwin
- b. **Ans** Lamarck

- c. Gregor
- d. Schwann

A random change in DNA that alters genetic information is

- a. Genetic drift
- b. Gene flow
- c. **Ans** Mutation
- d. Linkage

Appendix absent in humans but present and which functions as caecum in herbivores is termed

- a. Vegetative
- b. **Ans** Vestigial
- c. Extinct
- d. Propagative

The disease associated with the formation of abnormal haemoglobin S is

- a. Haemophilia
- b. Down syndrome
- c. **Ans** Sickle cell anaemia
- d. Malaria

The dark-coloured dorsal and light-coloured ventral surfaces in fishes are an example of

- a. Disruptive colouration
- b. Warning colouration
- c. Mimicry
- d. **Ans** Counter shading

The visible appearance of the characters of an organism is

- a. **Ans** Phenotype
- b. Genotype
- c. Allele
- d. Heterozygote

In a cross between heterozygous tall plant (Tt) and homozygous short plant (tt), the probability of tall to short plant in the F₁ generation is

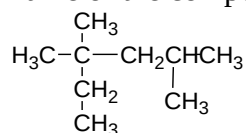
- a. 70:30
- b. **Ans** 50:50
- c. 90:10
- d. 80:20

The disease associated with the failure of blood to clot is

- a. Phenylketonuria
- b. Sickle cell anaemia
- c. Klinefelter
- d. **Ans** Haemophilia

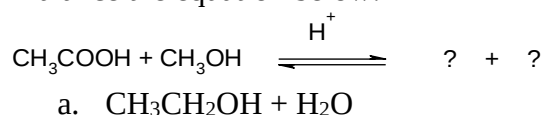
Chemistry

What is the IUPAC name of the compound below?



- a. 2-ethyl-2,4-dimethylpentane
- b. 4-ethyl-2,4-dimethylpentane
- c. **Ans** 2,4,4-trimethylhexane
- d. 3,3,5-trimethylhexane

Balance the equation below:



- b. **Ans** $\text{HCOOCH}_3 + \text{H}_2\text{O}$
- c. $\text{CH}_3\text{COOCH}_3 + \text{CO}_2$
- d. $\text{CH}_3\text{COOCH}_3 + \text{H}_2\text{O}$

The correct formula of methanal is _____.

- a. CH_3COOH
- b. **Ans** HCHO
- c. CH_3CHO
- d. CH_3OH

_____ is often called the fourth state of matter.

- a. **Ans**Plasma
- b. Solid
- c. Liquid
- d. Gas

Carbon reacts with excess air to produce _____.

- a. CO
- b. **Ans** CO_2
- c. CO_3
- d. CO_4

Which of the following statements is NOT correct?

- a. The average kinetic energy of a gas is directly proportional to its temperature
- b. **Ans**At constant temperature, the volume of a gas increases as the pressure increases
- c. The pressure of a gas is inversely proportional to its volume
- d. The temperature of a gas is directly proportional to its volume

An element X forms the following compounds with chlorine; XCl_4 , XCl_3 , XCl_2 . This illustrates the

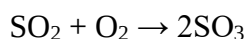
- a. **Ans**law of multiple proportions
- b. law of chemical proportions
- c. law of simple proportions
- d. law of conservation of mass
- e. law of definite proportions

A small quantity of solid ammonium chloride (NH_4Cl) was heated gently in a test tube, the solid gradually disappeared to produce two gases. Later, a white cloudy deposit was observed on the cooler part of the test tube. The ammonium chloride is said to have undergone distillation

- a. **Ans**sublimation
- b. precipitation
- c. evaporation
- d. decomposition

The sub-atomic particles located in the nucleus of an atom are?

- a. **Ans**neutron and proton
- b. proton and electron
- c. proton and ions
- d. neutron and electron



In the reaction above, the most suitable catalyst is?

- a. chromium(vi)oxide

- b. iron(iii)oxide
- c. copper(i)oxide
- d. **Ans**vanadium(v)oxide

2-methylprop-1-ene is an isomer of _____.

- a. 3-methyl but-1-ene
- b. 2-methyl but-1-ene
- c. **Ans**but-2-ene
- d. pent-2-ene

Elements X and Y have electronic configurations $1s^2 2s^2 2p^4$ and $1s^2 2s^2 2p^6 3s^2 3p^1$ respectively. When they combine, the formula of the compound formed is

- a. XY
- b. X₂Y
- c. **Ans**Y₂X₃
- d. X₂Y₂

By what amount must the temperature of 200cm³ of Nitrogen at 27°C be increased to double the pressure if the final volume is 150cm³ (Assume ideality)

- a. **Ans**150°C
- b. 177°C
- c. 75°C
- d. 120°C

Physic

An electric heater converts electrical energy 1KJ to heat energy every 2 seconds with 100 % efficiency. The power produced is

- a. **Ans** 0.5KW
- b. 2 KW
- c. 1KW
- d. 10 W

If a resistor is halved in magnitude and the potential across it is tripled, then the ratio of the old current to the new current is

- a. 1:3
- b. 3:1
- c. 6:1
- d. **Ans**1:6

The silver wall of a vacuum flask prevents heat loss due to

- a. Conduction
- b. Convection
- c. **Ans** Radiation
- d. Diffraction

A grasshopper jumps 1.00 m from rest, with an initial velocity at a 45° angle with respect to the horizontal. Find the maximum height reached.

- a. 2.000 m
- b. 0.125 m
- c. 0.500 m
- d. Ans 0.250m**

Three capacitors (2 μF , 3 μF , and 4 μF) are connected in parallel. Compute the equivalent capacitance.

- a. **Ans 0.92 μF**
- b. 9.00 μF
- c. 4.50 μF
- d. 100 μF

A conducting sphere has a net charge of $4.8 \times 10^{-7} \text{ C}$

- What is the approximate number of excess electrons on the sphere if the fundamental natural charge is $1.602 \times 10^{-19} \text{ C}$?
- a. 100
- b. Ans 300**
- c, 200
- d. 400

A battery has an internal resistance of 4 Ω and an emf of 12 V. The terminal voltage when a load of 20 is connected across it is

- a. 4V
- b. Ans 10 V**
- c. 20 V
- d. 2 V

An A.C. voltage is connected to an RLC series circuit of resistance, inductance, and capacitance of the circuit are 5 Ω , 3 H, and 0.05 F Calculate the resonance frequency.

- a. 1.10Hz
- b. 1.12Hz
- c. Ans 1.13 Hz**
- d. 1.14 Hz

The basic difference between the galvanometer and the electric motor is

- a. The size of the magnetic fields
- b. Ans The presence of hair springs in the galvanometer**
- c. The soft iron armature in the galvanometer
- d. The couple formed on the parallel sides of the rectangular coil in the galvanometer.

A small source emits sound waves with a power output of 80.0 W. Find the intensity of 3.00 m from the source.

- a. 2.670 W/m^2
- b. Ans 0.707 W/m^2**
- c. 8.90 W/m^2
- d. 0.790 W/m^2

The phenomenon which involves the increase the frequency of sound as the source approaches the observer and decrease as the source moves away from the observer is called

- a. Reverberation effect
- b. Echo Effect
- c. Ans Doppler Effect**
- d. Superposition effect

The potential difference across the ends of a wire is doubled in magnitude. If Ohm's law is obeyed, one of the following statements concerning the resistance of the wire is true?

- a. The resistance is one-half of its original value.
- b. The resistance is twice its original value.

- c. **Ans** The resistance is not changed
 d. The resistance increases by a factor of four.

1. The angle between the earth's magnetic field and the horizontal is called the angle of
 a. Declination
 b. **Ans** Dip
 c. Variation
 d. Inclination

Mathematics

The variance of x , $2x$, $3x$, $4x$ and $5x$ is

- a. $x\sqrt{2}$
 b. **Ans** $2x^2$
 c. x^2
 d. $3x$

Find the sum of the range and the mode of the set of numbers 10, 9, 10, 9, 8, 7, 7, 10, 8, 10, 8, 4, 6, 9, 10, 9, 10, 9, 7, 10, 6, 5.

- a. **Ans** 16
 b. 14
 c. 12
 d. 10

In how many ways can a delegation of 3 be chosen from among 5 men and 3 women, if at least one man and at least one woman must be included?

- a. 15
 b. 28
 c. 30
 d. **Ans** 45

A function $f(x)$ passes through the origin and its first derivative is $3x + 2$. What is $f(x)$?

- a. **Ans** $y = (3x^2)/2 + 2x$
 b. $y = (3x^2)/2 + x$
 c. $y = 3x^2 + (x/2)$
 d. $y = 3x^2 + 2x$

The expression $ax^2 + bx + c$ equals 5 at $x = 1$. If its derivative is $2x + 1$, what are the values of a , b , c respectively?

- a. 1, 3, 1
 b. 1, 2, 1
 c. 2, 1, 1
 d. **Ans** 1, 1, 3

Evaluate $21.05347 - 1.6324 \times 0.43$ to 3 decimal places.

- a. 20.980

- b. 20.351
- c. 20.981
- d. **Ans** 20.352

Simplify $\sqrt[3]{64a^3}^{-1}$

- a. 4a
- b. $1/8a$
- c. 8a
- d. **Ans** $1/4a$

A car dealer bought a second-hand car for N250,000 and spent N70,000 refurbishing it. He then sold the car for N400,000. What is the percentage gain?

- a. 60%
- b. 32%
- c. **Ans** 25%
- d. 20%

Find the principal which amounts to N5,500 at a simple interest in 5 years at 2% per annum.

- a. N4,900
- b. **Ans** N5,000
- c. N4,700
- d. N4,800

Evaluate $\frac{(0.14^2 \times 0.275)}{7(0.02)}$ to 3 decimal places.

- a. **Ans** 0.039
- b. 0.385
- c. 0.033
- d. 0.038

Divide: $a^{3x} - 26a^{2x} + 156a^x - 216$ by $a^{2x} - 24a^x + 108$

- a. **Ans** $a^x - 2$
- b. $a^x + 2$
- c. $a^x - 8$
- d. $a^x - 6$

If two graphs $y = px^2 + q$ and $y = 2x^2 - 1$ intersect at $x = 2$, find the value of p in terms q.

- a. $\frac{q-8}{7}$
- b. **Ans** $\frac{7-q}{4}$
- c. $\frac{8-q}{2}$
- d. $\frac{q+7}{8}$

Find the integral values of x and y satisfying the inequality $3y + 5x \leq 15$, given that $y > 0$, $y < 3$ and $x > 0$.

- a. (1,1), (1,2), (1,3)

- b. (1,1), (2,1), (1,3)
- c. (1,1), (3,1), (2,2)
- d. **Ans** (1,1), (1,2), (2,1)

English

In each of questions, choose the option that has the same vowel sound as the one represented by the letters) underlined.

Air

- a. Hear
- b. her
- c. fire
- d. **Ans** heir

Burn

- a. sum
- b. cousin
- c. **Ans** skirt
- d. port

In each of questions , choose the option that has the same consonant sound as represented by the letters) underlined.

Joke

- a. begs
- b. Bridged
- c. **Ans** scissors
- d. killed

Business

- a. grace
- b. rice
- c. **Ans** Eyes
- d. sink

Church

- a. **Ans**feature
- b. Chauffeur
- c. ocean
- d. machine

Choose the option that is nearest in meaning to the word underlined:

In this matter, we must be prepared to tolerate human frailties.

- a. quistiveness
- b. **Ans** weaknesses
- c. profligacy
- d. innovation.

I detest his lackadaisical attitude.

- a. **Ans** carefree
- b. lazy
- c. supercilious

- d. disloyal.

Nobody will endure such profligate spending habits.

- a. generous
- b. arrogant
- c. **Ans** wasteful
- d. mindless.

Choose the option that is most nearly opposite in meaning to the underlined word and that will, at the same time, correctly fill the gap in the sentence

The management has neither..... nor denied the rumour that it was going to lay off some workers.

- a. Pronounced
- b. **Ans** confirmed
- c. rejected
- d. advertised.

Instead of..... the issue, the speaker digressed to unimportant matters.

- a. summarizing
- b. describing
- c. revealing
- d. **Ans** addressing.

The telephone booths were all.... and I had to wait for more than ten minutes for one to become vacant.

- a. locked
- b. busy
- c. **Ans** occupied
- d. ringing

Biology

One of these can be classified as a living or non-living thing

- a. Bacteria
- b. Fungi
- c. **Ans** Virus
- d. Algae

Heterotrophic mode of life with extracellular digestion is a feature of

- a. Lichen
- b. **Ans** Fungi
- c. Algae
- d. Cyanobacteria

Lack of structures to represent the leaf, stem and root are unique to

- a. Liverwort
- b. Moss
- c. **Ans** Algae
- d. Mycorrhiza

Which of these transports water and mineral salts in pteridophytes

- a. Phloem

- b. Rhizoid
- c. Root
- d. **Ans** Xylem

Seed-producing plants are collectively called

- a. **Ans** Gymnosperm
- b. Spermatophyte
- c. Angiosperm
- d. Pteridophytes

Which of the following shows the highest evolutionary advancement

- a. Moss
- b. Fern
- c. Spirogyra
- d. **Ans** Pine

An example of a filamentous alga is

- a. Chlorella
- b. Chlamydomonas
- c. Ulva
- d. **Ans** Spirogyra

The common bread mould is

- a. Penicillium
- b. Agaricus
- c. Aspergillus
- d. **Ans** Rhizopus

The cell wall of one of the following contains chitin

- a. Monocots
- b. Bacteria
- c. **Ans** Penicillium
- d. Volvox

The structure that absorbs nutrients from digested organic matter in fungi is

- a. Sporangiphore
- b. Sporangium
- c. **Ans** Rhizoid
- d. Spore

The most specialized protozoan in the following is

- a. **Ans** Paramecium
- b. Euglena
- c. Amoeba
- d. Trypanosome

Bilharziasis is caused by

- a. Plasmodium
- b. Escherichia coli
- c. **Ans** Schistosoma
- d. Retrovirus

Regulation of excess water in Paramecium is achieved by

- a. Food vacuole
- b. **Ans** Contractile vacuole
- c. Gullet
- d. Trichocyst

Chemistry

The catalyst used in esterification reaction is _____.

- a. concentrated $C_6H_{12}O_6$
- b. concentrated HN_3
- c. concentrated CH_3Cl
- d. **Ans** concentrated H_2SO_4

a. Which of the following does not exist?

Ans methene

- b. methanal
- c. butanone
- d. propanal

Chlorine is used as a bleaching agent. The bleaching action is due to _____.

- a. **Ans** oxidation
- b. chlorination
- c. hydrogenation
- d. reduction

Oxygen is absent in?

- a. **Ans** Kerosene
- b. Soil
- c. Glass
- d. Cement

Soaps clean surfaces on the principle based on?

- a. viscosity
- b. floatation
- c. elasticity
- d. **Ans** surface tension

Calculate the solubility in mol dm^{-3} of 40g of $CuSO_4$ dissolved in 100g of water at 120°C .
[Cu = 64, S = 32, O = 16]

- a. 4.00
- b. **Ans** 2.50
- c. 0.40
- d. 0.25

Coffee stains can best be removed by?

- a. kerosine
- b. turpentine
- c. **Ans** a solution of borax in water
- d. ammonia solution

Which of the following salts is slightly soluble in water?

- a. $AgCl$

- b. **Ans**CaSO₄
- c. Na₂CO₃
- d. PbCl₂

Iron can be protected from corrosion by coating the surface with?

- a. gold
- b. silver
- c. copper
- d. **Ans**zinc

Which of the following is used in rocket fuels?

- a. **Ans**HNO₃
- b. CH₃COOH
- c. H₂SO₄
- d. HCl

A constituent common to bronze and solder is?

- a. lead
- b. silver
- c. copper
- d. **Ans**tin

Stainless steel is used for making?

- a. magnets
- b. **Ans**tools
- c.coins and medals
- d. moving parts of clocks

Which of the following is used as fuel in miners' lamps?

- a. Ethanal
- b. **Ans**Ethyne
- c. Ethene
- d. Ethane

Physics

A light-sensitive area at the back of the eye where images are formed is called

- a. yellow Spot
- b. Iris
- c. Ciliary Muscle
- d. **Ans** Retina

A luminous object is placed at a given distance from a converging lens of focal length 12 cm and a real image is produced. Find the image distance if it is equal to the object distance.

- a. 6 cm
- b. 3 cm
- c. 12 cm
- d. **Ans** 24 cm

The lower and upper fixed points on a thermometer are 40 and 120 respectively. Its reading at 60° C is

- a. 60
- b. 40

- c. 160
- d. **Ans** 88

The temperature in an electric iron is regulated by

- a. Thermometer
- b. **Ans** Bimetallic strip
- c. Steel
- d. Conductor

A bead traveling on a straight wire is brought to rest at 0.2 m by friction. If the mass of the bead is 0.01 kg and the coefficient of friction between the bead and the wire is 0.1, determine the work done by friction

- a. $2 \times 10^{-4} \text{J}$
- b. **Ans** $2 \times 10^{-3} \text{J}$
- c. $2 \times 10^1 \text{J}$
- d. $2 \times 10^2 \text{J}$

A machine whose efficiency is 60 % has a velocity ratio of 5. If a force of 500 N is applied to lift a load of P (N), what is the magnitude of P?

- a. 750 N
- b. 4166 N
- c. 500 N
- d. **Ans** 1500 N

A motor tyre is inflated to a pressure of $2.0 \times 10^5 \text{ Nm}^{-2}$ when the temperature of the air is 27°C . What will be the pressure at 87°C assuming the volume does not change?

- a. $2.6 \times 10^{-5} \text{ Nm}^{-2}$
- b. **Ans** $2.4 \times 10^{-5} \text{ Nm}^{-2}$
- c. $2.2 \times 10^{-5} \text{ Nm}^{-2}$
- d. $1.3 \times 10^{-5} \text{ Nm}^{-2}$

A beam of light is incident from air to water at an angle of 30° . Find the angle of refraction if the refractive index of water is $4/3$.

- a. 15°
- b. 18°
- c. **Ans** 22°
- d. 24°

Determine the inductive reactance when a 30 mH inductor with negligible resistance is connected to a 1.3 KHz oscillator

- a. 39.0Ω
- b. 122.0Ω
- c. **Ans** 245.0Ω
- d. $39.0 \text{ K } \Omega$

The half-life of a radioactive element is 9 days. Calculate the fraction that remains after 36 days.

- a. $1/32$
- b. **Ans** $1/16$
- c. $1/4$
- d. $1/8$

The graphite rods surrounding the uranium fuel rods in a nuclear reactor are used to

- a. Absorb the neutrons and hence halt the nuclear process
- b. Create the neutrons- and hence slow down the nuclear process

- c. **Ans** Slow down the neutrons and hence slow down the nuclear process
- d. Speed up the neutrons and hence speed up the nuclear process

The force on a current-carrying conductor in a magnetic field is greatest when the

- a. Conductor makes an angle of 60° with the field
- b. Force is independent of the angle between the field and the conductor
- c. Conductor is parallel with the field
- d. **Ans** Conductor is at right angles with the field

Electrical power is transmitted at a high voltage rather than a low voltage because the amount of energy loss is due to

- a. **Ans** Heat dissipation
- b. Production of Eddy current
- c. Excessive current discharge
- d. Excessive voltage discharge

Mathematics

Factorize completely $x^2 + 2xy + y^2 + 3x + 3y - 18$

- a. **Ans** $(x + y + 6)(x + y - 3)$
- b. $(x - y - 6)(x - y + 3)$
- c. $(x - y + 6)(x - y - 3)$
- d. $(x - y - 6)(x + y + 3)$

Tope bought X oranges at N5.00 each and some mangoes at N4.00 each. If she bought twice as many mangoes as oranges and spent at least N65.00 and at most N130.00, find the range of values of X .

- a. $4 \leq X \leq 5$
- b. $5 \leq X \leq 8$
- c. **Ans** $5 \leq X \leq 10$
- d. $8 \leq X \leq 10$

Three consecutive positive integers k , l and m are such that $l^2 = 3(k+m)$. Find the value of m .

- a. 4
- b. 5
- c. 6
- d. **Ans** 7

The first term of a geometric progression is twice its common ratio. Find the sum of the first two terms of the G.P if its sum to infinity is 8.

- a. $\frac{8}{5}$
- b. $\frac{8}{3}$
- c. **Ans** $\frac{72}{25}$

d. $56/9$

Divide $4X^3 - 3x + 1$ by $2x - 1$

- a. $2X^2 - X + 1$
- b. $2X^2 - X - 1$
- c. $2X^2 + X + 1$
- d. **Ans** $2X^2 + X - 1$

Find a positive value of α if the coordinate of the Centre of a circle $X^2 + Y^2 - 2\alpha x + 4y - \alpha = 0$ is $(\alpha, -2)$ and the radius is 4 units.

- a. 1
- b. 2
- c. **Ans** 3
- d. 4

A man 1.7m tall observes a bird on top of a tree at an angle of 30° . if the distance between the man's head and the bird is 25m, what is the height of the tree?

- a. 27.6m
- b. **Ans** 14.2m
- c. $1.7 + (25 \frac{\sqrt{3}}{3})$ m
- d. $1.7 + (25 \frac{\sqrt{2}}{2})$ m

Express $\frac{1}{x^3-1}$ in partial fractions

- a. **Ans** $\frac{1}{3}(\frac{1}{x-1} - \frac{(x+2)}{x^2+x+1})$
- b. $\frac{1}{3}(\frac{1}{x-1} - \frac{x-2}{x^2+x+1})$
- c. $\frac{1}{3}(\frac{1}{x-1} - \frac{(x-2)}{x^2+x+1})$
- d. $\frac{1}{3}(\frac{1}{x-1} - \frac{(x+1)}{x^2-x-1})$

In $\triangle MNO$, $MN = 6$ units, $MO = 4$ units and $NO = 12$ units. If the bisector of $\angle M$ meets NO at P , calculate NP .

- a. 4.8 Units
- b. **Ans** 7.2 Units
- c. 8.0 Units
- d. 18.0 Units

Find the tangent to the acute angle between the lines $2x + y = 3$ and $3x - 2y =$

- a. $-7/4$
- b. $7/8$

c. **Ans** 7/4

d. 7/2

From a point P, the bearings of two points Q and R are $N67^{\circ}W$ and $N23^{\circ}E$ respectively. If the bearing of R from Q is $N68^{\circ}E$ and $PQ = 150m$, calculate PR.

a. 120m

b. 140m

c. **Ans** 150m

d. 160m

Find the equation of the locus of a point $P(x,y)$ such that $PV = PW$, where $V = (1,1)$ and $W = (3,5)$

a. $2x + 2y = 9$

b. $2x + 3y = 8$

c. $2x + y = 9$

d. **Ans** $x + 2y = 8$

Find the area bounded by the curve $y = x(2-x)$. The x-axis, $x = 0$ and $x = 2$.

a. 4 sq units

b. 2 sq units

c. **Ans** $\frac{4}{3}$ sq units

d. $\frac{1}{3}$ sq units

UNIVERSITY OF MEDICAL SCIENCES, ONDO
POST UNIFIED TERTIARY MATRICULATION SCREENING
2024

ENGLISH LANGUAGE

In each of questions BELOW, choose the word or group of words that best completes each of the sentences.

The principal has already summoned all the on the matter.

a. **Ans** heads of departments

b. head of department

c. heads of department

d. head of department

..... Science and Technology, many African countries have not made significant progress since their independence.

a. with regards to

b. in respect to

c. as far as

d. **Ans** with regard to

We have no choice _____ to do what they demand.

- a. except
- b. than
- c. unless
- d. **Ans**but

The committee has submitted its report _____ the students
in
for
about
Anson

Choose the option that is most nearly opposite in meaning to the underlined word and that will, at the same time, correctly fill the gap in the sentence

The management has neither..... nor denied the rumour that it was going to lay off some workers.

- a. Pronounced
- b. **Ans** confirmed
- c. rejected
- d. advertised.

Instead of..... the issue, the speaker digressed to unimportant matters.

- a. summarizing
- b. describing
- c. revealing
- d. **Ans** addressing.

The telephone booths were all.... and I had to wait for more than ten minutes for one to become vacant.

- a. locked
- b. busy
- c. **Ans** occupied
- d. ringing

The common practice among some media practitioners is to be sensational in their reporting.

- a. prevalent
- b. **Ans**rampant
- c. ordinary
- d. cogent

Choose the option that is nearest in meaning to the word underlined:

In this matter, we must be prepared to tolerate human frailties.

- a. quisitiveness
- b. **Ans** weaknesses
- c. profligacy
- d. innovation.

A bank clerk should be able to tell easily the difference between.... Notes and counterfeit ones.

- a. crisp
- b. good
- c, correct
- c. **Ans**genuine.

That fateful decision changed the company's outlook in many ways.

- a. wonderful
- b. **Ans**disastrous
- c. uncontrollable
- d. unsuccessful

BIOLOGY

In a dicot leaf, guard cells differ from other epidermal cells because they

- a. have no definite shape
- b. lack nuclei
- c. are smaller
- d. **Ans**contain chloroplasts

Which of the following structures is NOT found in the female agama lizard?

- a. Pre-anal pads
- b. **Ans**Nuchal crest
- c. Gular fold
- d. Nasal scale

Beriberi is caused by a deficiency of vitamin

- a. **Ans**B
- b. C
- c. A
- d. E

If an isolated living cell is left in distilled water for two hours, it is likely to

- a. lose some of its water to the surrounding water
- b. lose all of its water to the surrounding water
- c. reproduce by binary fission
- d. **Ans**become more turgid.

Spirogyra and Mucor can be grouped together as Thallophyta because

- a. they are unicellular organism
- b. their spores could be dispersed by wind
- c. they are capable of living independent lives
- d. **Ans**their bodies are made up of thallus and filaments

The mammalian organ through which nourishment and oxygen diffuse into a developing embryo is called

- a. **Ans**Placenta
- b. chorion
- c. umbilical cord
- d. oviduct

The intermediate product in the anaerobic breakdown of glucose in fungi and germinating seeds is

- a. Citric acid
- b. Lactic acid
- c. **Ans** Pyruvic acid
- d. Glycolic acid

Plants that can survive in an environment where water is scarce are

- a. **Ans** Xerophytes
- b. Hydrophytes
- c. Mesophytes
- d. Thermophytes

A short thick beak adapted for crushing seeds is a feature of

- a. Sunbird
- b. **Ans** Sparrow
- c. Eagle
- d. Duck

A random change in DNA that alters genetic information is

- a. Genetic drift
- b. Gene flow
- c. **Ans** Mutation
- d. Linkage

Appendix absent in humans but present and which functions as caecum in herbivores is termed

- a. Vegetative
- b. **Ans** Vestigial
- c. Extinct
- d. Propagative

The hypha of rhizopus is said to be non-septate because it

- a. Does not contain chlorophyll
- b. **Ans** Has no cross wall
- c. Is vacuolated
- d. Stores oil globules

The most common characteristic of the fungal hyphae is the possession of

- a. **Ans** Cell-like compartment with one nucleus each
- b. Cell walls that are both rigid and chitinous
- c. A multicellular mycelium in the substrate
- d. Cell-like partitions formed by cross-walls

CHEMISTRY

Coffee stains can best be removed by?

- a. kerosene
- b. turpentine
- c. **Ans** a solution of borax in water
- d. ammonia solution

Which of the following salts is slightly soluble in water?

- a. AgCl

- b. **Ans**CaSO₄
- c. Na₂CO₃
- d. PbCl₂

Iron can be protected from corrosion by coating the surface with?

- a. gold
- b. silver
- c. copper
- d. **Ans**zinc

Calculate the molar concentration of HCl if 50 ml of HCl is required to neutralise 25 ml of 1 M NaOH in acid base titration.

- a. 0.25 M
- b. **Ans**0.5 M
- c. 1.0 M
- d. 2.0 M

Which of the following is a bleaching agent?

- a. **Ans**sodium hypochlorite
- b. sulphur dioxide
- c. liquid ammonia
- d. ether

The aluminium belongs to group _____ and period _____ in the periodic table.

- a. 5 and 4
- b. 5 and 2
- c. **Ans**3 and 3
- d. 7 and 0

Aluminium is extracted from _____ by _____.

- a. **Ans**alumina, electrolysis
- b. alum, electrolysis
- c. alumina, pyrolysis
- d. alum, photolysis

The shape of the water molecule is _____ shape.

- a. trigonal
- b. linear
- c. **Ans**V (bent)
- d. trigonal planar

The type of water which produces large amount of lather with soap is known as _____ water.

- a. **Ans** soft
- b. hard
- c. amorphous
- d. lime

The type of hardness that is due to the presence of chlorides and sulphates of calcium and magnesium is _____ hardness.

- a. temporary
- b. **Ans**permanent
- c. amorphous

d. gel

Carbon dioxide contains carbon and oxygen in a fixed ratio 3:8 by mass. Name the law of chemical combination that governs this statement.

- a. **Ans** law of constant proportion
- b. Avogadro's Law
- c. Boyle's law
- d. Gay-Lussac Law

The correct formula of methanal is _____.

- a. CH_3COOH
- b. **Ans** HCHO
- c. CH_3CHO
- d. CH_3OH

_____ is often called the fourth state of matter.

- a. **Ans** Plasma
- b. Solid
- c. Liquid
- d. Gas

PHYSICS

A conducting sphere has a net charge of $4.8 \times 10^{-7} \text{ C}$. What is the approximate number of excess electrons on the sphere if the fundamental natural charge is $1.602 \times 10^{-19} \text{ C}$?

- a. 100
- b. **Ans** 300
- c. 200
- d. 400

A battery has an internal resistance of 4Ω and an emf of 12 V. The terminal voltage when a load of 20 is connected across it is

- a. 4V
- b. **Ans** 10 V
- c. 20 V
- d. 2 V

An A.C. voltage is connected to an RLC series circuit of resistance, inductance, and capacitance of the circuit are 5Ω , 3 H, and 0.05 F. Calculate the resonance frequency.

- a. 1.10 Hz
- b. 1.12 Hz
- c. **Ans** 1.13 Hz
- d. 1.14 Hz

Which of the following statements are true about the spring balance and the chemical balance? I. Both are

used to measure the mass of an object. II. Either of them may be used to measure the weight of an object. III. The spring balance principle of moments. IV. A change in gravity changes the readings of a spring balance but not that of a chemical balance.

- a. I and IV
- b. II and III
- c. **Ans**III and IV
- d. I, II, and III

Which of the following units is equivalent to kgms^{-1} ?

- a. Ns^{-1}
- b. Nms
- c. **Ans** Ns
- d. Js^{-1}

A football is hit with a velocity of 6ms^{-1} at an angle of 30° to the horizontal. The time of flight of the ball is

- a. 0.6s
- b. 5s
- c. **Ans**0.3s
- d. 3s

The image of an object is located 6cm behind a convex mirror. If its magnification is 0.6, calculate the focal length of the mirror

- a. 3.75cm
- b. 6.60cm
- c. **Ans**15.00cm
- d. 30.0 cm.

When a ray of light passes from the water to glass, it is

- a. bent away from the normal
- b. **Ans**bent towards the normal
- c. totally internally reflected
- d. absorbed by the glass

A 45° triangular glass prism can be used as a reflector of light because

- a. refraction never takes place in such a prism
- b. the angle of reflection equals the angle of incidence
- c. the refractive index of glass is less than 1
- d. **Ans**the critical angle for glass is less than 45° .

The optical phenomenon responsible for rainbow is

- a. refraction
- b. diffusion
- c.reflection
- d. **Ans**Dispersion

Why is it impossible for the lines of force of an electric field to cross one another

- a. like charges repel each other
 - b. a charged particle is deflected
 - c. like changes are arranged on the same lines of force
 - d. **Ans**a positively charged particle can travel in only one direction at any time
- none of these is correct.

Suppose a layer of oil with $n = 1.50$ coats the surface of the water. What is the critical angle for total internal reflection for light traveling in the oil layer and encountering the oil-water boundary?

- a. 47.7°
- b. 49.7°
- c. 59.7°
- d. **Ans** 67.7°

Which of the following statements is correct

- a. Uniform speed always implies uniform velocity
- b. Uniform speed always implies zero acceleration
- c. **Ans** Uniform speed may imply nonzero acceleration
- d. Constant momentum implies constant acceleration

ENGLISH LANGUAGE

Choose the option that is most nearly opposite in meaning to the underlined word and that will, at the same time, correctly fill the gap in the sentence

The arrival of the police at the scene..... rather than mitigate tension

- a. Provoked
- b. **Ans** heightened
- c. created
- d. prolonged.

The management has neither _____ nor denied the rumour that it was going to lay off some workers.

- a. pronounced
- b. **Ans** confirmed
- c. rejected
- d. advertised

Choose the option that is nearest in meaning to the word underlined:

I detest his lackadaisical attitude.

- a. **Ans** carefree
- b. lazy
- c. supercilious
- d. disloyal.

Roses are common in May but.... in November.

- a. dead
- b. strange
- c. **Ans** rare
- d. extinct.

Choose from the options the word that has the same vowel sound in italics as the one represented by

the letter(s)

Rust

- a. loud
- b. touch
- c. bought
- d. **Ans**march

Research

- disturb
- comfort
- Ans**affair
- carry

Choose the word that best complete the following sentences

If the jury's verdict is "not guilty," the defendant will be _____

- a. **Ans**acquitted
- b. indicted
- c. paroned
- d. forgiven

Choose the interpretation that you consider most appropriate for this sentences:

Ladies and gentlemen, the worse may be over, but we are not out of the woods yet. This means that we are

- a. still in the forest
- b. not yet ready to celebrate our success
- c. **Ans**likely to encounter more difficulties
- d. not sure of ourselves

Choose the word that best completes the following sentence:

Surprisingly, Musa turned the job offer

- a. away
- b. in
- c. out
- d. **Ans**down

Choose the word or group of words that best completes the following sentence:

We have no choice to do what they demanded.

- a. except
- b. than
- c. unless
- d. **Ans**but

Choose the option that is nearest in meaning to the word underlined:

In this matter, we must be prepared to tolerate human frailty.

- a. inquisitiveness
- b. **Ans**weakness
- c. profligacy
- d. innovation.

Choose the option that is nearest in meaning to the word underlined:

I detest his lackadaisical attitude.

- a. **Ans**carefree
- b. lazy
- c. supercilious

d. disloyal

Choose the option that is opposite in meaning to the word underlined.

Peter always approaches issues optimistically.

- a. realistically
- b. **Ans** pessimistically
- c. carelessly
- d. unrealistically

Choose the option that is opposite in meaning to the word underlined.

Why should Dixon be evasive in answering the question?

- a. indirect
- b. **Ans** direct
- c. guessing
- d. sure

BIOLOGY

Muscles fatigue is caused by

- a. Accumulation of ethanol in the muscle cells
- b. Accumulation of methanol in the muscle cells
- c. **Ans** Accumulation of lactic acid in the muscle cells
- d. Accumulation of pyruvic acid in the muscle

The group of insects that have mouth parts adapted for both piercing and sucking is

- a. Cockroach, aphids and mosquitoes
- b. Aphids, houseflies and moths
- c. **Ans** Mosquitoes, tsetse flies and aphids
- d. Aphids, beetles and grasshoppers

Which of the following does a virus have in common with animal cells?

- a. Nucleus
- b. **Ans** DNA
- c. Glycogen
- d. Cytoplasm

A random change in DNA that alters genetic information is

- a. Genetic drift
- b. Gene flow
- c. **Ans** Mutation
- d. Linkage

Seed-producing plants are collectively called

- a. **Ans** Gymnosperm
- b. Spermatophyte
- c. Angiosperm
- d. Pteridophytes

Which of the following shows the highest evolutionary advancement

- a. Moss
- b. Fern
- c. Spirogyra
- d. **Ans** Pine

An example of a filamentous alga is

- a. Chlorella
- b. Chlamydomonas
- c. Ulva
- d. **Ans** Spirogyra

Goiter is often associated with the deficiency of

- a. Chlorine
- b. Calcium
- c. Copper
- d. **Ans** Iodine

OST One of these is not a fat-soluble vitamin

- a. Vitamin A
- b. **Ans** Vitamin C
- c. Vitamin E
- d. Vitamin K

Benedict's solution is used to test for

- a. Starch
- b. Proteins
- c. Fats
- d. **Ans** Reducing sugars

Which of the following insects lays its eggs in horny, purse-like cases?

- a. Mosquito
- b. Butterfly
- c. Termite
- d. **Ans** Cockroach

A stage of mitosis during which sister chromatids are pulled apart to opposite ends of the cell is

- a. Interphase
- b. Prophase
- c. Metaphase
- d. **Ans** Anaphase

The process by which liquid materials are taken up into the cell is termed

- a. **Ans** Pinocytosis
- b. Phagocytosis
- c. Endocytosis
- d. Exocytosis

CHEMISTRY

Reaction of methanal with NaBH_4 gives _____.

- a. methane
- b. **Ans** methanol
- c. ethanol
- d. propanone

Which substance is called a universal solvent?

- a. hexane
- b. **Ans**water
- c. ethanol
- d. glucose

Which of the following are the correct allotropes of sulphur?

- a. **Ans**yellow rhombic sulphur (α -sulphur) and the monoclinic (β -sulphur)
- b. yellow rhombic sulphur (α -sulphur) and the tetraclinic (δ -sulphur).
- c. green rhombic sulphur (γ -sulphur) and the monoclinic (β -sulphur).
- d. black rhombic sulphur (μ -sulphur) and the monoclinic (β -sulphur).

The catalyst used in esterification reaction is _____.

- a. concentrated $C_6H_{12}O_6$
- b. concentrated HN_3
- c. concentrated CH_3Cl
- d. **Ans**concentrated H_2SO_4

Ethanoic acid is a weak acid because?

- a. it totally dissociates into its constituent ions when dissolved in water
- b. it does not dissociate into its constituent ions when dissolved in water
- c. **Ans**It only partially dissociates into its constituent ions when dissolved in water
- d. it completely dissociates into its constituent ions when dissolved in water

Faraday's constant is _____ Coulomb mole⁻¹

- a. **Ans**96485
- b. 96500
- c. 9650
- d. 965

Methyl orange is _____.

AnsPink in acidic medium, yellow in basic medium
Yellow in acidic medium, pink in basic medium
Colourless in acidic medium, pink in basic medium
Pink in acidic medium, colourless in basic medium.

Lime water is _____.

- a. CaO
- b. **Ans** $Ca(OH)_2$
- c. $CaCO_3$
- d. $CaCl_2$

The summation of pH and pOH is equal to _____

- a. 10
- b. 7
- c. **Ans**14
- d. 12

The chemical formula of caustic potash is _____.

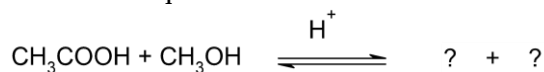
- a. $NaOH$

- b. $\text{Ca}(\text{OH})_2$
- c. NH_4OH
- d. **Ans** KOH

2-methylprop-1-ene is an isomer of _____.

- a. 3-methyl but-1-ene
- b. 2-methyl but-1-ene
- c. **Ans** but-2-ene
- d. pent-2-ene

Balance the equation below:



- a. $\text{CH}_3\text{CH}_2\text{OH} + \text{H}_2\text{O}$
- b. **Ans** $\text{HCOOCH}_3 + \text{H}_2\text{O}$
- c. $\text{CH}_3\text{COOCH}_3 + \text{CO}_2$
- d. $\text{CH}_3\text{COOCH}_3 + \text{H}_2\text{O}$

By what amount must the temperature of 200cm³ of Nitrogen at 27°C be increased to double the pressure if the final volume is 150cm³ (Assume ideality)

- a. **Ans** 150°C
- b. 177°C
- c. 75°C
- d. 120°C

PHYSICS

A quantity is defined as the product of cross-sectional area and change in momentum per unit length. Which of the following is its SI unit?

- a. Nm
- b. Kgs^{-1}
- c. **Ans** $\text{Kgm}^2\text{s}^{-2}$
- d. NKgs^{-1}

A piece of copper wire of length 100m at 30 °C is heated to 120 °C. If its linear expansivity is $1.8 \times 10^{-4} / \text{K}$, its new length is

- a. 166.6 cm
- b. **Ans** 101.6 cm
- c. 108.6 cm
- d. 1.08 cm

What happens to the total energy of a moving object if all the applied forces acting on the object are conserved?

- a. Increased
- b. Decrease

- c. **Ans** Remain constant
- d. Unstable

Which one of the following statements concerning the magnetic force on a charged particle in a magnetic field is true?

- a. It is a maximum if the particle is stationary.
- b. It is a maximum if the particle moves parallel to the field.
- c. It acts in the direction of motion for a positively charged particle.
- d. **Ans** It depends on the component of the particle's velocity that is perpendicular to the field

A ball of mass 200 g moving with a velocity of 4m/s collides and sticks with another ball of mass 300 g moving in the same direction with velocity 4m/s. What is the common velocity of the balls after the collision?

- a. **Ans** 5.6 m/s
- b. 2.8 m/s
- c. 11.2 m/s
- d. 1.4 m/s

An electric heater converts electrical energy 1KJ to heat energy every 2 seconds with 100 % efficiency. The power produced is

- Ans** 0.5KW
- 2 KW
- 1KW
- 10 W

A conducting sphere has a net charge of $4.8 \times 10^{-7} \text{ C}$

What is the approximate number of excess electrons on the sphere if the fundamental natural charge is $1.602 \times 10^{-19} \text{ C}$?

- a. 100
- b. **Ans** 300
- c. 200
- d. 400

A battery has an internal resistance of 4Ω and an emf of 12 V. The terminal voltage when a load of 20 is connected across it is

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- b. **Ans** 10 V
- c. 20 V
- d. 2 V

An A.C. voltage is connected to an RLC series circuit of resistance, inductance, and capacitance of the circuit are 5Ω , 3 H, and 0.05 F Calculate the resonance frequency.

- a. 1.10Hz
- b. 1.12Hz
- c. **Ans** 1.13 Hz
- d. 1.14 Hz

A small source emits sound waves with a power output of 80.0 W. Find the intensity of 3.00 m from the source.

- a. 2.670 W/m²

- b. **Ans** 0.707 W/m^2
- c. 8.90 W/m^2
- d. 0.790 W/m^2

The phenomenon which involves the increase the frequency of sound as the source approaches the observer and decrease as the source moves away from the observer is called

- a. Reverberation effect
- b. Echo Effect
- c. **Ans** Doppler Effect
- d. Superposition effect

A 1000 kg elevator is descending vertically with an acceleration of 1.0 ms^{-2} , if the acceleration due to gravity is 10.0 ms^{-2} , the tension in the suspending cable is

- a. 1.0 N
- b. **Ans** $9\ 000.0 \text{ N}$
- c. $11\ 000.0 \text{ N}$
- d. 9.0 N

When taking a penalty kick, a footballer applies a force of 30.0 N for a period of 0.05 s . If the mass of the ball is 0.075 kg , calculate the speed with which the ball moves off.

- a. 4.50 ms^{-1}
- b. 11.25 ms^{-1}
- c. **Ans** 20.00 ms^{-1}
- d. 45.00 ms^{-1}

UNIVERSITY OF MEDICAL SCIENCES, ONDO
POST UNIFIED TERTIARY MATRICULATION SCREENING
2025

MATHEMATICS

The angle sum of a triangle is?

- a. 90°
- b. 270°
- c. **Ans** 180°
- d. 360°

The circumference of a circle with radius 7 cm is?

- a. 21π
- b. **Ans** 14π
- c. 49π
- d. 7π

The distance between points (1, 2) and (4,6) is:

- a. 3
- b. 4
- c. **Ans**5
- d. 6

Find the limit as $x \rightarrow 0$ of the function $\frac{\sin x}{x}$

- a. 0
- b. Undefined
- c. ∞
- d. **Ans**1

The critical point of $f(x) = x^2 - 4x + 5$ is at $x =$

- a. 0
- b. **Ans**2
- c. 4
- d. 5

What does b represent in a regression line $y = a + bx$?

- a. Intercept
- b. **Ans**Slope

- c. Residual
- d. Error

The correlation coefficient r ranges from

- a. 0 to 1
- b. 1 to 100
- c. $-\infty$ to ∞
- d. **Ans**-1 to 1

How many 4-digit PINs can be formed using the digits 1, 2, 3, 4 if repetition is allowed?

- a. 24
- b. **Ans**256
- c. 64
- d. 128

A test for a disease is 95% accurate. If 2% of the population has the disease, what is the probability that a person who tests positive actually had the disease?

- a. 2%
- b. **Ans**28%
- c. 50%
- d. 95%

A vaccine is 80% effective. If 10 people take it, what is the probability that exactly 8 are protected?

- a. 0.5
- b. 0.403
- c. **Ans**0.302
- d. 0.504

A hospital is 8 km North of a clinic. A pharmacy is 6 km East of the clinic. What is the bearing of the pharmacy from the hospital?

- a. **Ans**36.87°
- b. 6.8°
- c. 53.13°
- d. 90°

A hospital floor is a rectangle measuring 30m by 20m. What is the area?

- a. 50 m²
- b. 60 m²

- c. 500 m²
- d. **Ans**600 m²

ENGLISH

Choose from the options, the word synonymous with the one underlined: He has been called a seasoned administrator because of the decision.

- a. bad
- b. wise
- c. **Ans**experienced
- d. aggressive

Choose from the options, the word synonymous with the one underlined: Such lackadaisical attitude is abominable.

- a. **Ans**careless
- b. enthusiastic
- c. unfriendly
- d. empathetic

Identify the phonetic symbol signifying the underlined sound: Cattle

- a. /θ/
- b. /ð/
- c. **Ans**/l/
- d. /dʒ/

Identify the phonetic symbol signifying the underlined sound: Choir

- a. /ð/
- b. /tʃ/
- c. /ʃ/
- d. **Ans**/k/

“All hands must be on deck to curb crime in our society”. The figure of speech used in the preceding sentence is

- a. Metonymy
- b. Hyperbole
- c. **Ans**Synecdoche
- d. Personification

“Death, you are shameless.” The preceding expression in inverted comma is an example of

- a. Innuendo
- b. **Ans** Apostrophe
- c. Hyperbole

d. Metaphor

Choose the right options to fill the missing gap: Many a pastor _____ in tongues.

- a. **Ans** speaks
- b. speak
- c. speaking
- d. spoked

Choose the right options to fill the missing gap: I want to buy _____

- a. **Ans**spaghetti
- b. spaghetti
- c. supergeti
- d. supergetti

Choose the option that best conveys the meaning of the underlined portion in the sentence:

There is some obvious symmetry in the whole presentation

- a. confusion
- b. hesitation
- c. excitement
- d. **Ans**orderliness

Choose the option that best conveys the meaning of the underlined portion in the sentence:

His jail terms were to run concurrently

- a. **Ans** simultaneously
- b. uniformly
- c. laboriously
- d. consecutively

Choose the option that is most opposite in meaning to the underlined word.

The bill has to wait as we are now insolvent

- a. bankrupt
- b. unsettled
- c. insoluble
- d. **Ans**affluent

BIOLOGY

In classification, the largest taxonomic group is:

- a. Family
- b. Order
- c. **Ans** Kingdom
- d. Genus

The part of a seed that develops into the root is the:

- a. Plumule
- b. **Ans** Radicle
- c. Cotyledon
- d. Endosperm

Which of the following plant hormones promotes the ripening of fruits?

- a. Auxin
- b. Cytokinin
- c. Gibberellin
- d. **Ans** Ethylene

The role of an organism in its environment is known as its:

- a. Habitat
- b. **Ans** Niche
- c. Territory
- d. Adaptation

The energy flow in an ecosystem is:

- a. Cyclic
- b. **Ans** Unidirectional
- c. Reversible
- d. Bidirectional

Which of the following is a density-dependent factor?

- a. Drought
- b. **Ans** Predation
- c. Flood
- d. Fire

Which of these is a macronutrient required by plants?

- a. Boron
- b. Manganese

- c. **Ans** Potassium
- d. Zinc

Which enzyme digests protein in the stomach?

- a. Amylase
- b. **Ans** Pepsin
- c. Lipase
- d. Maltase

The main site of absorption of digested food in humans is:

- a. Mouth
- b. Stomach
- c. **Ans** Small intestine
- d. Large intestine

Which of the following foods is rich in carbohydrates?

- a. Fish
- b. **Ans** Rice
- c. Egg
- d. Milk

Night blindness results from a deficiency of:

- a. **Ans** Vitamin A
- b. Vitamin B
- c. Vitamin C
- d. Vitamin E

Which of the following excretory products is the least toxic?

- a. **Ans** Uric acid
- b. Ammonia
- c. Urea
- d. Carbon dioxide

The part of the ear responsible for balance is the:

- a. Cochlea
- b. Tympanum
- c. Ossicles
- d. **Ans** Semi-circular canals

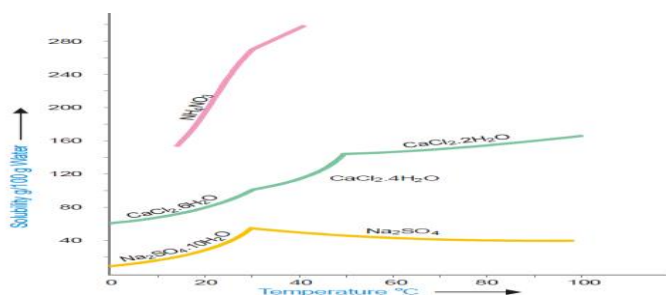
CHEMISTRY

The correct formula of methanal is _____.

- a. CH_3COOH
- b. **Ans** HCHO
- c. CH_3CHO
- d. CH_3OH

The graph below is termed?

- a. Continuous Curves
- b. **Ans** Discontinuous Curves
- c. Temperature Curve
- d. Pressure Curve



In Ohm's law, resistance (R) is measured in _____.

- a. ampere
- b. volts
- c. **Ans** ohms
- d. all of the above

In 5.49 seconds, 1.20g of argon-35 decays to leave only 0.15g. What is the half-life?

- a. **Ans** 1.83 seconds
- b. 2.75 minutes
- c. 2.75 seconds
- d. 1.83 minutes

The pH of a solution of HCL is 4. This shows that the molarity of the solution is

- a. **Ans** 4.0M
- b. 0.4M
- c. 0.0001M
- d. 0.001M

The sub-atomic particles located in the nucleus of an atom are?

- a. **Ans** neutron and proton
- b. proton and electron
- c. proton and ions
- d. neutron and electron

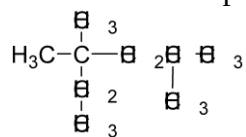
2-methylprop-1-ene is an isomer of _____.

- a. 3-methyl but-1-ene
- b. 2-methyl but-1-ene
- c. **Ans** but-2-ene
- d. pent-2-ene

If acidified Potassium dichromate (VI) ($K_2Cr_2O_7$) acts as oxidizing agent, colour changes from _____.

- a. yellow to green
- b. **Ans** orange to green
- c. yellow to red
- d. orange to red

What is the IUPAC name of the compound below?



- a. 2-ethyl-2,4-dimethylpentane
- b. 4-ethyl-2,4-dimethylpentane
- c. **Ans** 2,4,4-trimethylhexane
- d. 3,3,5-trimethylhexane

The constituent common to duralumin and alnico is _____.

- a. Co
- b. Mn
- c. **Ans** Al
- d. Mg

Faraday's constant is Coulomb mole⁻¹

- a. **Ans** 96485
- b. 96500
- c. 9650
- d. 965

Lime water is _____.

- a. CaO
- b. **Ans** Ca(OH)₂
- c. CaCO₃
- d. CaCl₂

Helium nuclei particles are called _____.

- a. Gamma particles
- b. Beta particles
- c. **Ans** Alpha particles
- d. No particles that are helium nuclei.

PHYSICS

The dimensions of work and energy are

- a. **Ans** MLT^{-3}
- b. MLT
- c. ML^2T^{-2}
- d. ML^{-3}

Two particle X and Y starting from rest cover the same distance. The acceleration of X is twice that of Y. The ratio of the time taken by X to that taken by Y is

- a. $\frac{1}{2}$
- b. **Ans** 2
- c. $\frac{1}{\sqrt{2}}$
- d. $\sqrt{2}$

Which of the following statements are correct about an object in equilibrium under parallel forces?

I. The total clockwise moments of the forces about any point equals the total anticlockwise moments about the same point.

II. The total forces in one direction equal the total forces in the opposite direction

III. The resolved components along the x-axis equal the resolved components along the y-axis

- a. **Ans** I and II only
- b. I and III only
- c. II and III only
- d. I, II and III

Which of the following electromagnetic waves has the highest frequency

- a. infrared rays
- b. **Ans** X-rays
- c. Ultra-violent rays
- d. radio waves

The water at the bottom of a waterfall is slightly warmer than that at the top because part of the potential energy raises the water temperature

water absorbs and heat from the atmosphere

Ans kinetic energy is changed into internal energy

dropping mass of water increases the heat capacity

A and C above

Steam from boiling water causes more damage to the skin than does boiling water because

- a. **Ans** water has a high specific heat
- b. steam has latent heat of fusion
- c. the steam is at a higher temperature than the water
- d. steam brings heat more easily by convection.

The small droplets of water that form on the grass in early hours of the morning is

- a. hail
- b. mist
- c. **Ans** dew
- d. fog

At low humidity in an environment, the human skin is usually

- a. dry and smooth
- b. damp and smooth
- c. **Ans** damp and rough
- d. dry and rough.

A boat at anchor is rocked by waves whose crests are 100 m apart and whose velocity is 25 ms^{-1} .

At what interval does the wave crest reach the boat?

- a. 2,500s
- b. 75s
- c. 4s
- d. **Ans** 0.25s

MATHEMATICS

If two tangents are drawn from an external point to a circle, they are?

- a. **Ans** Equal in length
- b. Parallel

- c. Perpendicular
- d. Collinear

In a cyclic quadrilateral, opposite angles sum to?

- a. 90^0
- b. **Ans** 180^0
- c. 270^0
- d. 360^0

If $n(A) = 10$, $n(B) = 8$, and $n(A \cap B) = 3$, then $n(A \cup B)$ is?

- a. 8
- b. 10
- c. 18
- d. **Ans** 15

If $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$ Find $A \cup B$

- a. $\{3\}$
- b. $\{3, 4, 5\}$
- c. **Ans** $\{1, 2, 3, 4, 5\}$
- d. $\emptyset$

For sets P and Q, $P \setminus Q$ represents?

- a. Elements in both P and Q
- b. **Ans** Elements in P but not in Q
- c. Elements in Q but not in P
- d. All elements in P and Q

An ambulance drives 10 km East, then turns 60° North-East and drives another 10 km. How far is it from the starting point?

- a. **Ans** 17.32 km
- b. 10 km
- c. 20 km
- d. 14.14 km

A circular hospital garden has a radius of 14m. What is its circumference?

- a. 44 m
- b. **Ans** 88 m
- c. 154 m
- d. 616 m

A doctor has 5 vaccines to administer to 3 patients, with each patient getting at least one. How many ways can this be done?

- a. 15
- b. 30
- c. 120
- d. **Ans** 150

The mean recovery time for a disease is 14 days, with a standard deviation of 2 days. What range covers 68% of patients (assuming normal distribution)?

- a. 10-18 days
- b. **Ans** 12-16 days
- c. 8-20 days
- d. 14-16 days

In a clinical trial, 75% of patients improved with Treatment A, while 60% improved with Treatment B. If 100 patients took each treatment, what is the difference in success rates?

- a. 5%
- b. 25%
- c. **Ans** 15%
- d. 35%

A hospital spends ₦500,000 monthly on salaries and ₦200,000 on drugs. If the total budget is ₦1,000,000, what percentage is left for other expenses?

- a. 50%
- b. 10%
- c. 20%
- d. **Ans** 30%

If 3 packs of medicine cost ₦1,200, how much will 7 packs cost?

- a. ₦2,400
- b. **Ans** ₦2,800
- c. ₦3,600
- d. ₦4,200

In a class, 60% of students are girls, and 30% of girls wear glasses. What is the probability a randomly chosen student is a girl who wears glasses?

- a. 0.3
- b. 1
- Ans** 0.18
- 0.90

ENGLISH

Identify the phonetic symbol signifying the underlined sound: harm

- a. /ʃ/
- b. /s/
- c. **Ans**/a:/
- d. /h/

Select the option that best explains the information conveyed in the sentence:

The events of last Friday show that there is no love lost between the Principal and the Vice-Principal.

- a. They like each other
- b. They work independently
- c. They cannot part company
- d. **Ans** They dislike each other

Select the option that best explains the information conveyed in the sentence:

The politician's inflammatory remarks were heavily criticised.

- a. The politician's rude remarks were heavily criticised
- b. **Ans** The politician's provocative remarks were heavily criticised
- c. The politician's bombastic remarks were heavily criticised
- d. The politician's tribalistic remarks were heavily criticized

Select the option that best explains the information conveyed in the sentence:

Oche's chief idiosyncrasy is a passion for pounded yam.

- a. Oche's chief hates pounded yam
- b. Oche hates pounded yam
- c. **Ans** Oche has a penchant for pounded yam
- d. Oche's chief likes pounded yam

Select the option that best explains the information conveyed in the sentence:

You must not attend the end-of-year party.

- a. **Ans** It is not necessary that you attend the party
- b. It is necessary that you do not attend the party
- c. You do not have to decide whether to attend the party or not
- d. You have to decide whether to attend the party or not

Choose the correct options that fit into the gap:

The girl _____ hair was cut is crying.

- a. who
- b. whom

- c. **Ans** whose
- d. which

Choose the correct options that fit into the gap:

- a. The angry man said he had been taken for a _____.
full
- b. **Ans** fool
- c. food
- d. foul

Choose the option that is most opposite in meaning to the underlined word.

The plaintiff made very cogent submissions to the trial judge

- a. bailiff
- b. prosecutor
- c. lawyer
- d. **Ans** defendant

Choose the option that is most opposite in meaning to the underlined word.

The long straight trunk of the tree is ideal for roundwood used like railway sleepers, refters and fence posts

- a. perfect
- b. **Ans** most unsuitable
- c. satisfactory
- d. poor

Choose the option that best conveys the meaning of the underlined portion in the sentence:

The bill has to wait as we are now insolvent

- a. overworked
- b. **Ans** bankrupt
- c. unsettled
- d. insoluble

Choose the option that best conveys the meaning of the underlined portion in the sentence:

All his plans fell through

- a. **Ans** failed
- b. were accomplished
- c. had to be reviewed
- d. were rejected

BIOLOGY

Which of the following is a modified stem?

- a. **Ans** Onion bulb

- b. Sweet potato
- c. Carrot
- d. Cassava

Which of the following is a C4 plant?

- a. Rice
- b. **Ans** Maize
- c. Wheat
- d. Barley

Which of the following is NOT a product of anaerobic respiration in yeast?

Which enzyme is secreted in saliva?

- a. Maltase
- b. **Ans** Amylase
- c. Lipase
- d. Trypsin

Which of these organisms is used in the production of cheese?

- a. **Ans** Streptococcus lactis
- b. Salmonella typhi
- c. Bacillus anthracis
- d. Vibrio cholerae

The causative agent of cholera is:

Salmonella

Ans Vibrio cholerae

Clostridium

Shigella

Which of the following is NOT a method of food preservation?

- a. Salting
- b. Refrigeration
- c. dFermentation
- d. **Ans** Irrigation

Which of the following parasites causes river blindness?

- a. Trypanosoma
- b. **Ans** Onchocerca volvulus
- c. Plasmodium
- d. Schistosoma

The study of the relationship between organisms and their environment is:

- a. Anatomy
- b. **Ans** Ecology
- c. Genetics

d. Morphology

Organisms that feed on dead and decaying organic matter are called:

- a. Parasites
- b. **Ans** Saprophytes
- c. Autotrophs
- d. Carnivores

Which biome is characterized by very low rainfall and succulents?

- a. **Ans** Desert
- b. Grassland
- c. Rainforest
- d. Tundra

The greenhouse effect is caused mainly by the accumulation of:

- a. Oxygen
- b. Nitrogen
- c. **Ans** Carbon dioxide
- d. Hydrogen

The gradual replacement of one community by another in an ecosystem is called:

- a. Competition
- b. **Ans** Succession
- c. Adaptation
- d. Stratification

CHEMISTRY

A sample of any of the following substances disappears after some time when exposed to air like camphor, naphthalene, or dry ice. This phenomenon is called?

- a. **Ans** Sublimation
- b. Evaporation
- c. Diffusion
- d. Radiation
- e.

Carbon-carbon double bonds consist of _____ and _____.

- a. **Ans** 1 sigma (δ) bond and 1 pi (π) bond
- b. 2 sigma (δ) bond and 1 pi (π) bond
- c. 1 sigma (δ) bond and 2 pi (π) bond
- d. 2 sigma (δ) bond and 2 pi (π) bond

In stable organic compounds, carbon will always form _____ bond.

- a. 2
- b. **Ans** 4

- c. 3
- d. 5

Lipids consist of _____ or _____.

- a. fat and glucose
- b. oil and fructose
- c. **Ans** fat and oil
- d. glucose and glucose

A hydrocarbon is found to contain 81.80% carbon and 18.20% hydrogen. Its empirical formula will be _____.

- a. C_2H_6
- b. **Ans** CH_3
- c. C_4H_8
- d. C_3H_6

Calculate the molar concentration of HCl if 50 ml of HCl is required to neutralise 25 ml of 1 M NaOH in acid base titration.

- a. 0.25 M
- b. **Ans** 0.5 M
- c. M
- d. M

Which of the following is a bleaching agent?

- a. **Ans** sodium hypochlorite
- b. sulphur dioxide
- c. liquid ammonia
- d. ether

The aluminium belongs to group _____ and period _____ in the period table.

- a. 5 and 4
- b. 5 and 2
- c. **Ans** 3 and 3
- d. 7 and 0

Aluminium is extracted from _____ by _____.

- a. **Ans** alumina, electrolysis
- b. alum, electrolysis
- c. alumina, pyrolysis
- d. alum, photolysis

Calculate the oxidation number of sulphur in H_2SO_4 .

- a. +2
- b. +4

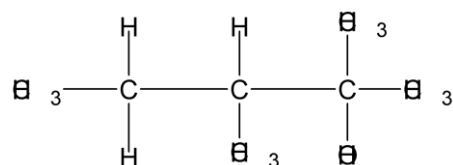
- c. **Ans** +6
- d. +8

Which of the following is an oxidizing agent used in preparation of carboxylic acids from aldehydes?

- a. **Ans** acidified KMnO_4
- b. acidified NaBH_4
- c. PCl_5
- d. concentrated H_2SO_4

Which of the following element is used in vulcanization of rubbers?

- a. sodium
- b. aluminium
- c. **Ans** sulphur
- d. potassium



The IUPAC name of the compound below is _____.

- a. **Ans** 2,3-dimethylpentan-2-ol
- b. 2,3-dimethylhexan-2-ol
- c. 3,4-dimethylpentan-2-ol
- d. 3,4-dimethylpentan-4-ol

PHYSICS

A weight of 1000 grams hangs from a lever 20cm to the right of the fulcrum. At the left is a 500 gram weight 20cm from the fulcrum. What is the value of x that will make the lever balanced?

- a. **Ans** 50cm
- b. 20cm
- c. 10cm
- d. 30cm
- e. 70cm.

The resultant of two forces acting on an object is maximum if the angle between them is

- a. 45°
- b. **Ans** 0°
- c. 90°
- d. 180°

A uniform meter rule is balanced on a fulcrum placed at the 35cm mark by suspending a mass 120g at the 10cm mark. Calculate the mass of the metre rule

- a. 60g
- b. 80g
- c. 120g
- d. **Ans** 200g

How many beta particle(s) are emitted in the radioactive decay of $^{198}_{79}\text{Au} \rightarrow ^{198}_{80}\text{Hg}$

- a. **Ans** 1
- b. 2
- c. 3
- d. 4

A $15\mu\text{F}$ capacitors are connected to a 240V and 50Hz a.c source. Calculate the reactance of the capacitors. ($\pi = 3.142$)

- a. 16Ω
- b. 68Ω
- c. 106Ω
- d. **Ans** 212Ω .

A miner hears an echo 1.26 seconds after blowing a whistle near a cliff (speed of sound in air = 330ms^{-1}).

- a. 104.0m
- b. **Ans** 207.9m
- c. 415.8m
- d. 523.8m

A particle in circular motion performs 30 oscillations in 6s. Its angular velocity's

- a. **Ans** $10\pi\text{rads}^{-1}$
- b. $5\pi\text{rads}^{-1}$
- c. $6\pi\text{rads}^{-1}$
- d. $5\pi\text{rads}^{-1}$

A satellite revolving around the Earth is kept in its orbit by

- a. centrifugal forces only
- b. centripetal forces only
- c. centripetal and frictional forces
- d. **Ans** centripetal and centrifugal forces

Which of the following does not describe the image formed by a plane mirror?

- a. erect
- b. laterally inverted
- c. same distance from the mirror as the object
- d. **Ans** magnified E. virtual

A wire of length 400cm is stretched between two fixed points. When plucked, its fundamental frequency is 150Hz. Calculate the speed of the wave produced.

- a. 250ms^{-1}
- b. 625ms^{-1}
- c. **Ans** 1200ms^{-1}
- d. 1500ms^{-1}

Calculate the total distance covered by a train before coming to rest if its initial speed is 30ms^{-1} with a constant retardation of 0.1m^{-2} .

- a. 5 500m
- b. **Ans** 4 500m
- c. 4 200m
- d. 3000m

A particle moves in a circular orbit of radius 0.02m. If the speed of the particle is 0.88ms^{-1} , calculate its frequency in cycles per second.

- a. 2.0
- b. **Ans** 7.0
- c. 8.8
- d. 14.0

Let Δx be the uncertainty in the measurement of position and Δp the uncertainty in the measurement of momentum. Then the uncertainty principle relation is given as

- a. $\Delta x. \Delta p = h$
- b. $\Delta x. \Delta p \leq h$
- c. **Ans** $\Delta x. \Delta p \geq \hbar$
- d. $\Delta x. \Delta p > h$